

Liquidity Mechanisms in Decentralized Finance: Design and Fragmentation in Illiquid Real-World Asset Markets

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Abstract

The tokenization of real-world assets (RWAs) links traditional assets to a blockchain-based trading infrastructure, but whether sustainable liquidity can be created remains unclear. We study liquidity formation in a fully transparent tokenized market for illiquid RWAs where three market mechanisms coexist: automated market makers (AMMs), bilateral peer-to-peer (P2P) marketplaces, and issuer-provided Buyback. Guided by reduced-form frameworks of execution cost and liquidity provision, we analyze 777,038 secondary-market trades and contemporaneous market liquidity-states from 2019 to 2025. We find that fragmentation across mechanisms generates persistent trader sorting and price dispersion. Price signals and executed volume concentrate on P2P. AMMs provide immediacy but are structurally fragile: thin liquidity and low turnover in illiquid RWAs weaken liquidity-provider incentives, resulting in shallow, largely issuer-supported depth. During market stress, execution costs increase differentially across mechanisms, consistent with mechanism-specific liquidity constraints. Our findings show that tokenization alone does not ensure liquid secondary markets for illiquid assets and that liquidity outcomes become mechanism-specific under low turnover.

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Keywords: Real-World Assets (RWA), Tokenization, Decentralized Finance (DeFi), Market Microstructure, Market Fragmentation

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1. Introduction

The tokenization of real-world assets (RWAs) changes not only how assets are represented, but how liquidity is created, allocated, and sustained. For traditionally illiquid, slow-moving, and heterogeneous assets, tokenization promises to create and extend secondary-market liquidity by enabling fractional ownership and continuous on-chain trading. At the same time, tokenization fragments potential liquidity across different execution technologies. In tokenized RWA markets, three fundamental liquidity mechanisms exist: automated market makers (AMMs), bilateral peer-to-peer (P2P) marketplaces, and issuer-provided buyback or redemption facilities (Buyback). These mechanisms differ sharply in pricing rules, depth formation, and execution certainty. While tokenization facilitates secondary markets in general, the central question is how the coexistence of heterogeneous liquidity mechanisms shapes trading and liquidity formation for tokenized illiquid RWAs.

In traditional finance, securitization has improved liquidity primarily for large, homogeneous, and frequently traded assets by aggregating demand and reducing participation frictions (O’Hara, 2003). For such assets, tokenization primarily reallocates existing liquidity by changing trading, settlement, and custody technology. By contrast, when tokenization is applied to private, heterogeneous assets such as real estate, private credit, or private equity, it attempts something more demanding: the endogenous creation of secondary-market liquidity for claims on slow-moving fundamentals (Case and Shiller, 1989; Williams, 1995; Whitaker and Kräussl, 2020; Goetzmann et al., 2021). In this setting, valuation opacity, search frictions, limited participation, and infrequent information updates remain binding constraints. While on-chain functionalities such as collateralization, borrowing, and transparent settlement can facilitate trading (Xin Li et al., 2024), liquidity remains a property of market design rather than technology alone. Tokenization may therefore reallocate liquidity for liquid underlyings, but for illiquid RWAs it constitutes an attempt at liquidity transformation that raises fundamental coordination and incentive-compatibility challenges (Cong et al., 2025b).

Fragmentation is central to this problem. Theory shows that dispersed trading across venues can improve welfare when no single platform aggregates information efficiently, but also that segmentation can impair execution quality when deterministic pricing interacts with thin liquidity and limited participation (Malamud and Rostek, 2017; Chen and Duffie, 2021; Aoyagi and Ito, 2025). Empirical evidence from Decentralized Finance (DeFi) documents persistent cross-venue price dispersion and slow synchronization even for highly liquid crypto-native assets (Makarov and Schoar, 2020; Barbon and Ranaldo, 2024). In tokenized markets

for illiquid RWAs, these forces are amplified. Trading frequency is low, participation is constrained by institutional frictions such as KYC-based whitelisting, and no single mechanism reliably aggregates information. In such environments, fragmentation does not merely redistribute order flow, but interacts with market design to determine whether liquidity can be financed and sustained at all.

This paper studies these issues in the context of RealT, one of the earliest and largest RWA tokenization platforms, focusing on real estate assets. The platform provides a rare empirical environment in which AMM, P2P, and Buyback mechanisms coexist under a uniform token design with full on-chain observability, providing systematic empirical evidence on the secondary-market trading of tokenized illiquid RWAs. This setting serves as a transparent laboratory for studying liquidity mechanisms that are commonly employed across tokenized RWA platforms, allowing direct measurement of mechanism-specific execution frictions, trader sorting behavior, price dispersion, and liquidity provision. While not all tokenized RWA markets share the same configuration, many employ a combination of the liquidity mechanisms analyzed in this study, and observing their coexistence within a single transparent environment provides a clean setting to isolate how execution technology shapes liquidity in fragmented secondary markets (see Appendix A.1). To our knowledge, RealT is the only platform for tokenized illiquid RWAs with a track record of more than five years that simultaneously features all three liquidity mechanisms and fully publicly accessible blockchain transaction data, making it uniquely suited for comprehensive empirical analysis. We reconstruct transaction-level quotes, offers, pool states, and execution outcomes across all mechanisms, yielding a comprehensive transaction-level dataset. We discipline the empirical analysis with a strictly positive, reduced-form execution-cost framework that captures the core technological differences across mechanisms, complemented by a simple framework of liquidity provision.

The key takeaway of this study is therefore structural. In illiquid RWAs, tokenization alone does not ensure liquid secondary markets. Our results suggest that when different execution mechanisms coexist, traders sort endogenously, and liquidity formation is constrained by market design and incentive compatibility. Liquidity outcomes become mechanism-specific under low turnover. AMMs provide continuous quotes but fail to scale when trading activity is low, while P2P markets intermediate economically meaningful trades and concentrate price signals, but still resemble digitized search markets.

2. Related literature and institutional details

This section reviews the literature and institutional setting relevant to our analyses.

2.1. Literature

We organize the literature around the core tensions raised in the introduction: tokenization, DeFi microstructure, and market fragmentation.

Tokenization. Distributed ledger technology enables a decentralized and transparent infrastructure for asset representation: digital ownership transfer, automated execution of contracts, and real-time settlement (Buterin, 2014; Cong and He, 2019; Harvey et al., 2025). These innovations underpin the expansion of DeFi (Schär, 2021), yet the effectiveness of blockchain-based trading venues in fostering trading and liquidity formation for tokenized illiquid RWAs remains uncertain. The inherent characteristics of RWAs further complicate liquidity formation. Unlike cryptocurrencies, which are native to the blockchain, fungible, and actively traded across global markets, RWAs can be heterogeneous, exhibit low turnover, and depend on the integration of off-chain data such as valuations, legal records, and income flows. This dependency introduces additional frictions through whitelisting and price oracles (Harvey and Rabetti, 2024; Cong et al., 2025c). However, first empirical evidence from tokenized markets indicates that these markets struggle with thin trading volumes, while showing early signs of maturation (Swinkels, 2023; Bergkamp et al., 2025). Recent evidence on tokenized equities shows that blockchain-based markets can incorporate value-relevant information (Cong et al., 2025a), while tokenization may not uniformly create liquidity but instead reallocates or transforms liquidity depending on asset characteristics (Cong et al., 2025b).

Despite a growing literature on tokenized assets, empirical research on secondary-market liquidity for RWAs remains sparse. Most studies focus on primary issuance mechanisms, such as Security Token Offerings (STOs), and their role in lowering investment barriers through fractional ownership (Lambert et al., 2022; Kreppmeier et al., 2023). These studies leave limited evidence on how liquidity forms and how execution outcomes differ across trading mechanisms once assets begin trading among investors.

DeFi microstructure. Conventional centralized exchanges aggregate liquidity through limit order books (Kyle, 1985; O’Hara, 2003), complemented by dealer-style venues like over-the-counter (OTC) systems for less liquid markets (Duffie et al., 2005). Blockchain-based decentralized exchanges implement analogous functions through smart contracts. AMMs

are the dominant DeFi exchange design. They use algorithmic pricing to provide continuous on-chain quotes but can sustain persistent price deviations under trading frictions and expose traders to front-running attacks (Malinova and Park, 2017; Park, 2023; Barbon and Ranaldo, 2024; Capponi and Jia, 2025). A complementary class of decentralized venues are on-chain P2P protocols that resemble OTC execution. Existing studies primarily examine liquidity determinants in cryptocurrency markets, both on centralized (Brauneis et al., 2021) and decentralized exchanges (Zhu et al., 2025). However, little research has addressed how these liquidity mechanisms operate and interact in tokenized RWA markets. The coexistence of multiple liquidity mechanisms introduces structural complexity that can distort pricing and execution outcomes, motivating further investigation as more traditional assets move on-chain into an increasingly complex DeFi ecosystem.

Market fragmentation. Building on canonical models of coexisting trading platforms (Pagano, 1989; Malamud and Rostek, 2017; Chen and Duffie, 2021; Aoyagi and Ito, 2025), we examine how fragmentation across venues shapes trading and liquidity formation. These models show that when investors optimally split orders, a finite degree of fragmentation can improve aggregate trading outcomes even as prices on individual venues become less informative. At the same time, excessive segmentation diffuses information and increases arbitrage exposure (Makarov and Schoar, 2020). Extending these insights, Rostek and Yoon (2025) show in a decentralized multi-asset setting that product design and cross-security inference interact with price impact, so market quality may still improve under limited fragmentation even as individual-venue depth declines. Prior research has largely focused on pricing and information aggregation in centralized trading mechanisms (Kyle, 1985; Hasbrouck, 1995; O’Hara, 2003) or decentralized mechanisms studied in isolation (Capponi et al., 2026). How price information transmits across fundamentally different market designs remains largely untested. In particular, there is limited evidence on whether deterministic pricing, bilateral search, and appraisal-based quoting can jointly facilitate trading and support liquidity formation.

Markets for tokenized illiquid RWAs provide a transparent setting in which these theoretical tensions can be tested empirically.

2.2. *Liquidity mechanisms and market design*

Three distinct liquidity mechanisms coexist in our setting, operating under fundamentally different designs: AMMs, P2P, and Buyback.

AMMs. Protocols such as Uniswap V2 ([Adams et al., 2020](#)) provide continuous two-sided liquidity at deterministic prices set by the constant-product market-making (CPMM) rule (see Appendix Figure [A.1](#)). Pool quotes are mechanically determined by reserve ratios, so price signals are fully embedded in the CPMM curve and adjust deterministically to order flow. Execution is immediate, but slippage increases convexly with trade size. Liquidity providers (LPs) earn fees proportional to volume and may incur impermanent loss when relative token prices move asymmetrically ([Capponi and Jia, 2021](#)). Our setting includes three constant-product AMMs: Uniswap V1, Uniswap V2, and, primarily, Levinswap, a Uniswap V2 fork deployed on the Gnosis blockchain (Gnosis), a compatible blockchain connected to Ethereum mainnet (Ethereum) (see Appendix Figure [A.2](#) for an interface example). For each RWA token, the issuer creates and seeds a single liquidity pool, predominantly quoted against the USDC stablecoin, to enable immediate trading and attract external liquidity providers. Following a successful STO, pools are initialized with fixed issuer-sponsored liquidity. As the underlying RWA tokens generate cash flows, mainly rental income, LPs receive the income stream associated with the tokens they contribute, plus a pro-rata share of the cash flow tied to the issuer-sponsored position, intended to incentivize participation beyond trading fees.

P2P. Marketplaces such as YAM and Swapcat support bilateral, posted-offer trading with user-defined prices and quantities (see Appendix Figure [A.3](#) for an illustrative example). They operate as decentralized posted-offer markets: participants list non-negotiable, take-it-or-leave-it buy or sell offers, creating observable bid and ask signals without automated matching. Execution is therefore stochastic and depends on search frictions, posted depth, and deviations from appraisal-based or AMM-based prices. This structure resembles decentralized OTC trading: it offers flexible pricing, transparent quoting, and support for larger or less time-sensitive trades. The buy side is typically thin and often hidden, creating the structural asymmetry that later shapes our empirical analyses. The corresponding P2P price signals are posted bid and ask quotes, closely resembling OTC-style indicative offers that reveal sellers' reservation values and the sparse, partially hidden buy-side demand. For details on the offer book interface, see Figure [A.4](#) in Appendix.

Buyback. The issuer offers investors the option to sell RWA tokens back at an appraisal-based valuation, resembling a fund-style redemption mechanism (see Figure [A.5](#)). Pricing is stable but infrequently updated, execution requires a deterministic waiting period, and liquidity is subject to maximum caps, for example \$2,000 per wallet per week in our setting. While this mechanism guarantees execution at a known price, it offers limited flexibility and

weak real-time price signals, making it most suitable for larger redemptions, simple execution needs, or traders prioritizing certainty over market-improved pricing. The Buyback price signal is the appraisal-anchored valuation, which is explicitly tied to the externally certified asset appraisal divided by the outstanding token supply, and serves as a reference point for secondary-market pricing. Importantly, access to all three secondary-market mechanisms is permissioned. Wallets and RWA tokens must be whitelisted through the issuer platform’s KYC process, and only whitelisted wallets can trade or provide liquidity in the corresponding RWA tokens. For more institutional details, see Appendix [A.1.1](#).

3. Framework for execution cost and liquidity provision across mechanisms

This section focuses on the key economic implications that discipline the empirical analyses, while the Appendix [A.2](#) provides a detailed exposition of the framework.

3.1. Execution cost framework

Trader sorting.

Each trader i , who wishes to place a buy or sell order for a quantity q of a certain RWA token is exposed to three sources of execution cost: (i) proportional fees, (ii) price impact (slippage), and (iii) execution delay. All mechanisms share a common reference price P .¹ Since every total-cost expression contains the factor $qP > 0$, we normalize costs by qP throughout and work with mechanism-specific normalized execution costs.

AMMs. Let L_A denote AMM liquidity (RWA token units). Under the constant-product rule, the post-trade price after a buy or sell of size q is

$$P_{\text{new}} = \begin{cases} P \left(\frac{L_A}{L_A - q} \right)^2, & \text{for a buy,} \\ P \left(\frac{L_A}{L_A + q} \right)^2, & \text{for a sell.} \end{cases} \quad (1)$$

Under the assumption that the trader’s average execution-price impact is approximated by one half of the marginal price change from P to P_{new} , the normalized AMM execution costs are therefore

$$\begin{aligned} c_A^{\text{buy}}(q) &= F_A + \frac{1}{2} \left[\left(\frac{L_A}{L_A - q} \right)^2 - 1 \right], \\ c_A^{\text{sell}}(q) &= F_A + \frac{1}{2} \left[1 - \left(\frac{L_A}{L_A + q} \right)^2 \right]. \end{aligned} \quad (2)$$

P2P with depth ($L_P > 0$). With posted offer depth L_P , execution incurs sub-quadratic price impact PI^+ (Almgren and Chriss, 2001):

$$c_P^{(+)}(q; L_P) = F_P + \frac{1}{2}PI^+\left(\frac{q}{L_P}\right)^\delta, \quad \delta \in (0, 1), \quad (3)$$

so impact scales as $q^{1+\delta}$ ("walking the book") and decreases with available depth L_P . All execution-cost expressions and routing conditions for the case of zero P2P depth ($L_P = 0$) are provided in Appendix A.2.

Buyback. Buyback execution occurs at a fixed relative discount F_B and deterministic waiting time τ_B . Delay enters linearly via the trader's individual urgency preference λ :

$$c_B = F_B + \lambda\tau_B. \quad (4)$$

Routing conditions. Traders choose the mechanism with the lowest normalized cost.

Sell orders. AMM is preferred to Buyback when

$$\lambda < \frac{F_A - F_B + \frac{1}{2} - \frac{1}{2}\left(\frac{L_A}{L_A+q}\right)^2}{\tau_B}. \quad (5)$$

With depth ($L_P > 0$), P2P is preferred to Buyback when

$$\lambda > \frac{F_P - \frac{1}{2}PI^+(q/L_P)^\delta - F_B}{\tau_B}. \quad (6)$$

Finally, P2P dominates AMM for $L_P > 0$,

$$1 - \left(\frac{L_A}{L_A+q}\right)^2 - PI^+\left(\frac{q}{L_P}\right)^\delta > 2F_P - 2F_A. \quad (7)$$

Formal derivations of these comparative statics and buy order routing conditions are provided in Appendix A.2.

3.2. Liquidity provision framework

P2P markets can in principle support voluntary market making, but when trading frequency is low, incentives to supply continuous bid and ask liquidity are weak, mirroring traditional OTC markets. By contrast, AMMs provide continuous quotes mechanically, but the depth of those quotes depends on liquidity supplied by LPs. Thus, LP incentives for external

liquidity provision are economically central. While AMMs are well studied for crypto-native assets, their viability for tokenized illiquid RWAs remains largely unexplored. We therefore focus on AMM liquidity provision and develop a simple framework that links external liquidity to turnover, fee revenue, and issuer-sponsored incentives. In each liquidity pool, the token issuer contributes L_R units of RWA tokens, paired with an equal-value amount of quote tokens (here a stablecoin) at reference price P . The issuer further subsidizes liquidity provision by transferring the cash-flow yield associated with L_R to external LPs, allocated pro rata by liquidity shares. An external LP then chooses additional liquidity L in RWA tokens, together with the matched quote amount, so that total RWA-side liquidity equals $L_R + L$. Over the investment horizon, the pool experiences a (ex ante unknown) total buy q_b and sell q_s volume of RWA tokens. We summarize the *net order imbalance* as

$$\Delta q := q_b - q_s, \quad (8)$$

which is positive (negative) when buy (sell) orders dominate. We define *total trading activity* as

$$Q := q_b + q_s, \quad (9)$$

capturing overall turnover in the pool over the investment horizon. Both Δq and Q must be estimated by the external LP when choosing how much liquidity to supply.

The external LP's profit over the horizon has four components: (i) own and prorated yield on the RWA tokens supplied to the pool; (ii) fee income from total trading volume; (iii) valuation changes arising from the constant-product pricing rule²; and (iv) the opportunity cost of capital required to finance the liquidity provision. Here, μ is the *yield* earned by holding the RWA tokens, and r is the *LP's cost of capital*. The term $a = \mu - 2r$ captures the net valuation effect of holding base tokens rather than investment alternatives. The linear coefficient b incorporates (i) the valuation effect of issuer seed liquidity, $2L_R(\mu - r)$, (ii) expected fee income QF_A from trading activity Q at AMM fee rate F_A , and (iii) the valuation effect of expected net order flow Δq . The constant term $c = \mu L_R^2$ reflects the cash-flow value of issuer-supplied liquidity.

Collecting terms and normalizing by the reference price P , expected profit for an external LP takes the form

$$\Pi(L) = \frac{aL^2 + bL + c}{L_R + L}, \quad (10)$$

where the coefficients collect distinct economic effects:

$$\begin{aligned} a &= \mu - 2r, \\ b &= 2L_R(\mu - r) + QF_A + \Delta q(2 - \frac{\mu}{2}), \\ c &= \mu L_R^2. \end{aligned} \tag{11}$$

For convenience, we define a *revenue intensity* term

$$\gamma := QF_A + \Delta q(2 - \frac{\mu}{2}), \tag{12}$$

which bundles the turnover-driven fee component and the valuation effects associated with directional order flow. Under the empirically relevant condition $2r > \mu$, the profit function $\Pi(L)$ is concave in L .³ If the revenue-intensity satisfies $\gamma > 0$, the optimal external liquidity is given by an expression which tends to be negative for smaller values of γ . More importantly, the full set of liquidity choices for which the external LP earns weakly positive profits for all $L \in [0, L']$ can be characterized by

$$\begin{aligned} L' &= \frac{2L_R(\mu - r) + \gamma}{2(2r - \mu)} + \\ &\quad \frac{\sqrt{4L_R^2r^2 + 4L_R(\mu - r)\gamma + \gamma^2}}{2(2r - \mu)}. \end{aligned} \tag{13}$$

The expression shows that AMM liquidity increases with the issuer's seed liquidity L_R and with expected revenue intensity γ . Because $\partial L' / \partial L_R > 0$, issuer-sponsored liquidity expands the set of $(Q, \Delta q)$ realizations under which external LPs can profitably supply liquidity. We provide full derivations of the equations in Appendix A.2.1.

3.3. Hypotheses

Based on the theoretical frameworks, the following hypotheses will guide our empirical analyses:

Hypothesis 1 (Sorting, fragmentation, and persistent price dispersion). *Traders endogenously sort across execution mechanisms according to trade size and execution priorities, fragmenting order flow and generating cross-mechanism price dispersion.*

Hypothesis 2 (Turnover-driven AMM liquidity fragility). *Low trading activity reduces the revenue intensity of AMM pools and shrinks the profitability for external LPs. When turnover is insufficient to cover opportunity costs, AMM liquidity remains shallow.*

Hypothesis 3 (State-dependent amplification under stress). *Under execution stress, the co-existence of heterogeneous execution mechanisms amplifies trading frictions by intensifying order-flow fragmentation, leading to systematically larger divergences in execution outcomes.*

4. Data and summary statistics

This section describes the data and presents summary statistics used in the empirical analyses.

4.1. Platform

We study RealT, a tokenization platform that issues and trades RWAs on public blockchains. RealT has operated since 2019 and, by October 2025, had tokenized approximately \$163 million worth of underlying assets, primarily rental residential real estate in the US and Latin America. Its long track record and advanced ecosystem featuring multiple secondary-market mechanisms make it a unique data-rich laboratory for studying secondary markets for tokenized illiquid RWAs. More importantly, because many other tokenization platforms rely on a combination of these same mechanism categories, our study is naturally transferable beyond the RealT setting (see Appendix A.1).

All issuances and secondary-market trades occur on Ethereum and Gnosis and are executed via smart contracts, ensuring full on-chain traceability and allowing us to reconstruct complete transaction-level panels from November 2019 to October 2025, comprising 777,038 trades, \$41.2 million in traded volume, and 9,202 unique wallet addresses. Figure 1 summarizes the platform’s aggregate market evolution.

[«Insert Figure 1»]

We enrich each secondary-market trade with contemporaneous, liquidity-relevant market states by reconstructing pool-level AMM reserves and liquidity-provider positions, on-chain P2P offer books with standing bid and ask inventories, and Buyback flows. This yields mechanism-specific panels matched directly to individual trades.

At the token level, we merge these data with detailed token- and asset-level fundamentals for 752 RWA tokens, combining time-invariant attributes (e.g., STO terms and property characteristics) with time-varying fundamentals (e.g., rental income and appraisal values). Mechanism overlap is pervasive: 97% of tokens trade on both P2P and AMMs, and 73% trade on all three mechanisms. We aggregate the resulting components into token- and pool-level panels at multiple frequencies for the empirical analyses. Appendix A.4 provides full details on database construction, validation procedures, and reproducibility.

4.2. Price signals and trades

Figure 2 illustrates, for a single representative tokenized RWA and its tokens, the core implications of our theoretical framework. Each liquidity mechanism generates distinct price signals and trade prices because execution costs, delay, and depth differ systematically across mechanisms. Even for an appraisal-anchored asset, price signals display substantial dispersion, and realized trade prices fluctuate widely over time.

[«Insert Figure 2»]

Buyback prices remain flat, reflecting infrequent revaluations. In contrast, AMM price signals vary because quotes are mechanically tied to liquidity-pool reserves: even small order imbalances push pool prices away from appraisal values or contemporaneous P2P offer prices, creating temporary mispricing and quasi-arbitrage opportunities (see Appendix A.5). With both offer-sides present, P2P price signals display a persistent bid–ask spread, with asks reflecting seller-posted offers and bids representing posted demand. On the buy-side, filled P2P bids are rare: P2P buy-side liquidity is mostly hidden rather than posted.

In the full sample, Table 1 documents large and systematic differences in trading activity across mechanisms, with highly skewed intraday trading. Overall, individual trade sizes are typically very small (mean 0.97 tokens; median 0.03), with economic volume concentrated in a narrow right tail: the median trade volume is \$1.30, while the maximum exceeds \$89,000. AMMs handle approximately 71% of all trades by count, P2P 27%, and Buyback only 2%, with nearly all activity settling on the low-cost Gnosis (99%). This aggregate skewness maps cleanly into sharp cross-mechanism differences. AMM trades are extremely numerous but very small (median 0.016 tokens; mean volume \$6.26), consistent with their role in servicing immediacy-driven, low-value trades. P2P trades are roughly an order of magnitude larger (median 1 token; mean volume \$166.87) and account for the majority of economic volume.

Buyback trades are infrequent but involve the largest average trade sizes. P2P trades clear at the highest average prices (\$56.42), while AMM prices exhibit the greatest dispersion, reflecting thin pools and episodic mispricing. Cumulatively, P2P accounts for \$34.7 million (84%) in total traded volume—an order of magnitude more than either AMMs (8%) or Buyback (7%).

[«Insert Table 1»]

In aggregate form, these patterns scale into clear market-level trends. As shown in Figures 3 and 4, AMMs dominate by *number of trades* but not by *trading volume*: they intermediate many small, immediacy-driven trades, whereas P2P venues execute far fewer but substantially larger trades. This separation reflects the underlying execution technologies. Beginning in late 2021, lower transaction costs on Gnosis and maturing P2P infrastructure led to a pronounced shift of volume toward P2P venues. By early 2022, P2P had become the dominant mechanism by traded USD volume, while AMMs persisted mainly for small, time-sensitive flows. When not suspended, the Buyback remained a stable but infrequent outlet for large redemptions.

[«Insert Figure 3 and Figure 4 »]

4.3. Descriptive statistics

Table 2 presents descriptive statistics at the intraday (transaction-level) and daily aggregation levels for the key variables used throughout our analyses and subsample constructions, including trading outcomes, mechanism-specific liquidity-states and measures, and market control variables. Overall, our data set covers 752 distinct tokenized RWAs, totaling approximately 3 million tokens available for trading.⁴

Trades and price dispersion. Roughly 40% of trades are purchases. Price signals across mechanisms differ substantially in dispersion. Buyback prices cluster tightly around appraisal values (mean = \$51.14). P2P bid–ask quotes bracket these appraisal values in a manner consistent with posted-offer trading. AMM-implied prices, by contrast, exhibit extremely high dispersion (SD = \$5,014) driven by near empty liquidity pools that mechanically generate outsized implied quotes when reserves are thin. Although median AMM prices align closely with P2P and Buyback prices, the mean is dominated by a small number of extreme outliers.

AMM and P2P liquidity. Liquidity provision differs markedly across mechanisms. Observed pool liquidity per side ranges from essentially empty (min $\approx$ \$0) to deep pools (max

= \$89,070), with central tendency around \$1,000 (median = \$847; mean = \$1,144, per side). AMM slippage is observed only when trades execute against the pool and is not imputed on AMM-inactive observations; conditional on AMM execution, slippage is modest on average (mean = -1.2%; median = -0.5%) but can become extreme when reserves are thin (min = -99.9%). LP fee revenues are correspondingly small for most trades (90th percentile = \$0.019; mean = \$0.019), reflecting the generally low dollar trade sizes settled on AMMs.

P2P liquidity is posted and therefore intermittent and not guaranteed. In the trade-matched sample, bid quotes are observed far less often than ask quotes ($N = 167,381$ vs. $635,475$), consistent with greater standing sell interest. Median posted depth is \$517 on the bid side and \$993 on the ask side, but the distribution is highly right-skewed, with occasional very large ask-side inventories (max = \$433,436). The median time on market from offer creation until execution (partial or full fill) is roughly 2 days, indicating relatively rapid turnover; however, P2P execution remains probabilistic relative to the AMM's immediate execution, and can take substantially longer (mean ≈ 18 days; 90th percentile ≈ 48 days).

Stress event variables. In the absence of a unique market-clearing reference price, we compare mechanisms using their implied execution prices for a fixed-size trade. *Price impact* is measured as the implied execution cost of a fixed-size trade and expressed in basis points. We evaluate price impact for a representative 1-token trade (\$50). For AMM trades, price impact is derived from the constant-product pricing rule using contemporaneous pool states. Let P^{spot} denote the AMM-implied spot price and L^{pool} the total liquidity pool value in USD. The implied base-token reserve is inferred as $X = L^{\text{pool}}/P^{\text{spot}}$. The price impact of executing a 1-token trade is computed as $10,000 \times \left(1 - \frac{X}{X+1}\right)$, expressed in basis points, which captures the mechanical slippage induced by moving along the constant-product curve and reflects the execution cost implied by available AMM depth at the time of the transaction. For P2P trades, price impact is defined relative to the contemporaneous posted ask quote. Specifically, impact is computed as $10,000 \times (P^{\text{exec}} - P^{\text{ask}})/P^{\text{ask}}$, where P^{ask} is the liquidity-weighted ask price prevailing at the time of execution. This measure captures the price concession required to execute against posted bilateral liquidity.

In the later phase of the maturing platform starting with the end of 2024, we are able to directly observe two-sided bilateral quotes in the P2P market and construct a *bid–ask spread* as an alternative measure of market liquidity. For each P2P observation with both a realistic bid and ask quote, we compute the bid–ask spread as the midpoint-normalized difference between the prevailing liquidity-weighted ask and bid prices. Let P^{ask} and P^{bid} denote

the contemporaneous P2P ask and bid quotes, respectively, and define the midpoint price as $p^{\text{mid}} = (p^{\text{ask}} + p^{\text{bid}})/2$. The bid–ask spread is then measured as $10,000 \times (p^{\text{ask}} - p^{\text{bid}})/p^{\text{mid}}$, expressed in basis points. This measure captures the instantaneous cost of immediacy in bilateral trading and reflects both quote competition and available depth on each side of the P2P market. Bid–ask spreads are computed only for observations with active two-sided quotes.

Temporary Restraining Order (TRO) affected is an indicator equal to one if the token involved in the trade is subject to a temporary restraining order, and zero otherwise. *Asset size* measures the total concurrent USD appraisal value of the underlying real estate asset associated with the trade.

Market control variables. Lastly, we include a set of market controls capturing token-level conditions, blockchain environment factors, and broader macroeconomic states that may affect execution and routing. These include short-horizon RWA price volatility, crypto-market conditions proxied by Ether returns and transaction fees, and macro-financial conditions measured by changes in US Treasury yields, the ADS business conditions index, and lagged house price growth.

[«Insert Table 2»]

5. Empirical analysis

This section examines trader sorting across mechanisms and price dispersion (Section 5.1), liquidity provision (Sections 5.2), and trading frictions under systemic stress (Section 5.3).

5.1. Trader sorting across AMM, P2P, and Buyback

To test the sorting implications of Hypothesis 1, we estimate transaction-level sorting frameworks of the form

$$\Pr(\text{Trade}_{i,t} \in m) = \beta_1^{(m)} \log(\text{TradeVolume}_{i,t}) + \mathbf{C}'_{i,t} \boldsymbol{\gamma}^{(m)} + \mathbf{L}'_{i,t} \boldsymbol{\delta}^{(m)} + \alpha_i + \delta_t + \varepsilon_{i,t}^{(m)}. \quad (14)$$

where $\Pr(\text{Trade}_{i,t} \in m)$ is an indicator equal to one if transaction i on day t executes via mechanism $m \in \{\text{AMM}, \text{P2P}, \text{Buyback}\}$. The key regressor $\log(\text{TradeVolume}_{i,t})$ captures the size of the executed order. The vector $\mathbf{C}_{i,t}$ contains transaction-day execution cost controls, including blockchain fees, AMM slippage, short-term volatility, valuation deviations, ETH returns, and Treasury yield changes. The vector $\mathbf{L}_{i,t}$ captures mechanism-specific liquidity:

AMM pool depth, P2P ask-side inventory, and the time-on-market of the active P2P listing. All regressions include token fixed effects (α_i) and year-month fixed effects (δ_t), and standard errors are clustered at the token level.

Table 3 reports six models. Columns 1–3 show baseline specifications that include only trade size and execution-cost controls. Columns 4–6 control these specifications with mechanism-specific liquidity.

[«Insert Table 3»]

Trade size is the dominant determinant of sorting. Across all specifications, trade size is the dominant determinant of mechanism choice. Larger trades are significantly less likely to execute via AMMs and more likely to route to P2P or Buyback. These sorting patterns directly reflect the mechanism-choice condition implied by Equation 3. Execution costs materially reallocate trades across mechanisms. Higher blockchain fees and higher slippage reduce AMM usage and increase routing toward P2P venues and Buyback, reflecting differences in fees, execution certainty, and price impact across mechanisms. Market-wide controls such as volatility, valuation deviations, and macro variables enter with intuitive signs but play a secondary role relative to trade size and execution frictions.

Mechanism-specific liquidity reinforces sorting patterns. Mechanism-specific liquidity significantly sharpens sorting patterns, greater AMM pool depth increases the probability of AMM execution and reduces routing to P2P and Buyback mechanisms. Conversely, deeper P2P ask-side inventory and longer time-on-market increase the likelihood of P2P execution, consistent with posted-offer trading and search frictions. Appendix A.5 shows that the observed routing patterns are not driven by the volume of quasi-arbitrage activity across mechanisms. Instead, trades sort across mechanisms in response to mechanism-specific execution costs and liquidity constraints that govern economically meaningful execution.

5.2. Liquidity provision across mechanisms

AMM liquidity provision. AMM pools provide continuously available two-sided quotes implied by the reserve ratio under the constant-product design. Liquidity is supplied through LP deposits into the pool’s token reserves and removed through withdrawals, which mechanically scale depth on both sides of the market. Table 4 (Panel A) summarizes AMM liquidity provision across pools, distinguishing issuer and external LP activity.

Across the full sample, we identify 275 unique external LPs, accounting for 4,597 deposits and 3,321 withdrawals. Issuer liquidity is concentrated at pool inception and remains largely

stable thereafter: the number of issuer deposits ($N = 954$) closely matches the number of pools created and underlying RWA issuances, indicating limited active liquidity management over the pool life cycle.⁵

[«Insert Table 4»]

Figure 5 traces the combined issuer and external liquidity of all three AMMs over time. The March 2021 migration from Ethereum to Gnosis and the shift from Uniswap to Levinswap coincided with a marked increase in external LP entry. However, this initial expansion was short-lived, as Levinswap relied on a non-stablecoin quote token that collapsed in 2021, imposing substantial losses on external LPs and triggering a sharp contraction in external liquidity. Over the same period, the P2P mechanism gains volume share (Figure 3), reducing AMM trading volume and LPs' expected fee revenue. External liquidity does not meaningfully recover thereafter, even after May 2022, when all newly created RWA pools were paired with USDC, reducing LPs' inventory risk. External participation begins to rise again toward the end of 2023 and remains elevated through subsequent issuer regime changes. Overall, issuer-sponsored liquidity anchors AMM depth throughout the sample.

[«Insert Figure 5»]

To empirically test these descriptive patterns of external liquidity provision and evaluate Hypothesis 2, we follow [Lehar and Parlour, 2025](#) and [Zhu et al., 2025](#) and proxy AMM liquidity with USD pool size, often referred to as total value locked (TVL), but focus on the external LP component. We estimate the following pool-week regression:

$$y_{i,t+1} = \beta_1 x_{i,t}^{\text{fee}} + \beta_2 x_{i,t}^{\text{iss}} + \beta_3 v_{i,t} + \beta_4 x_{i,t}^{\text{iss}} \times d_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}, \quad (15)$$

where i indexes pools and t indexes weeks. $y_{i,t+1}$ is the average external liquidity supplied to pool i in week $t+1$, measured in USD at prevailing pool prices. $x_{i,t}^{\text{fee}}$ denotes the total trading-fee revenue earned by pool i in week t , with fees fixed at 0.25% of volume and allocated pro rata across LPs. $x_{i,t}^{\text{iss}}$ is issuer-sponsored liquidity in pool i , valued at pool-implied prices and determining the subsidy to external LPs at the end of week t . $v_{i,t}$ captures pool-level price deviations associated with impermanent loss, measured as the standard deviation of log exchange-rate changes between the RWA token and the quote token, following [Capponi and](#)

Jia, 2021). The interaction term $x_{i,t}^{\text{iss}} \times d_{i,t}$ captures whether issuer-sponsored liquidity matters more for external liquidity provision when fee revenue is low (below its mean). We include pool fixed effects α_i and year-month fixed effects δ_t , which absorb time-invariant heterogeneity across the underlying RWAs and pool design choices (including the pool’s quote-token pairing) as well as common time-varying shocks (including platform-wide regime changes). Standard errors are clustered at the pool level. The summary statistics for the pool-week panel are presented in Table 4 (Panel B).

Table 5 presents our regression results. In the log-log specifications, coefficients can be interpreted as elasticities. For example, in the fixed-effects model (2), a 1% increase in trading-fee revenue at time t is associated with an approximately 0.66% increase in external liquidity at $t + 1$, holding other covariates and fixed effects constant. This indicates that higher trading-fee revenue predicts greater subsequent external liquidity, consistent with Hypothesis 2 and the view that fees constitute a primary component of LP profits. In contrast, higher token-price volatility is associated with lower external liquidity, consistent with greater potential losses and loss-versus-rebalancing risk (Milionis et al., 2024).

[«Insert Table 5»]

Issuer-sponsored liquidity is positively related to external liquidity. Economically, a larger issuer position increases overall depth, reduces expected slippage, and can attract more trading and fees. It can further serve as a commitment signal that increases perceived platform trustworthiness. Furthermore, it enlarges the subsidized income stream that external LPs receive through the issuer’s yield-generating RWA tokens in the pool. When trading fee revenue is low, as captured by the interaction term in the regression, external LPs rely more heavily on subsidies when endogenous trading fee income is insufficient. This pattern is consistent with the AMM liquidity provision framework’s comparative statics (Appendix A.2.1): since the derivative of L' with respect to L_R decreases with γ , increases in L_R induce larger liquidity responses in low-trading fee revenue regimes.

The overall result, especially the positive coefficient of the LP fee revenues, points to a feedback loop: limited trading constrains fee revenue, which in turn limits external liquidity provision, preventing the endogenous deepening typically observed in crypto-native AMMs. Consistent with this self-reinforcing dynamic, AMM trading activity declines over time (Figures 3 and 4), with trading migrating toward P2P execution.

P2P liquidity provision. P2P liquidity is sparse and episodic, consistent with a bilateral posted-offer market. While ask-side depth is persistently available, bid-side depth increases only temporarily, most notably in early 2025, and then recedes. P2P liquidity provision, which we define as concurrent bid and ask quotes posted by the same wallet address, is even rarer. We identify only 55 “two-sided quote providers.” Table 4 (Panel C) summarizes their activity and Figure 6 places it in the context of the full P2P offer book. Two-sided quoting is generally short-lived (median overlap of 9 days) and shallow (median two-sided depth of \$100), consistent with opportunistic, non-inventory-based liquidity provision rather than continuous market making. Of these two-sided quote providers, 12 also provide liquidity in AMMs, and the issuer is never among them. Taken together, these patterns suggest little strategic external liquidity provision in P2P and limited overlap in liquidity provision across mechanisms.

[«Insert Figure 6»]

Buyback. Buyback liquidity is deterministic and issuer-controlled. Weekly redemption caps per investor restrict total executable quantity, and Buyback does not generate continuous market-based price signals. We treat it as a separate liquidity mechanism at a stable appraisal-based valuation.

Integrated order-book reconstruction. We combine AMM depth, P2P offer-book depth, and Buyback capacity into a unified order-book representation. Figure 7 shows the aggregated depth across mechanisms. AMM liquidity appears as a smooth but issuer-driven layer; P2P liquidity is sparse and asymmetric; and Buyback provides a stable backstop that activates intermittently.

[«Insert Figure 7»]

Overall, RealT’s AMMs function less like a decentralized venue sustained by competitive liquidity provision and more like an issuer-maintained inventory facility. External LP participation exists but remains shallow relative to issuer-sponsored depth. While standard channels emphasized in crypto AMM studies, such as adverse selection, impermanent loss, and fee competition, also operate in this setting (Aquilina et al., 2024; Lehar et al., 2024), a key distinction is the persistent presence of a central initiator attempting to sustain external liquidity beyond launch. In most DeFi environments, AMM liquidity is initially seeded and then scales endogenously with trading volume and fee revenue, often supported by protocol-token incentives at launch (Heimbach et al., 2021; Capponi et al., 2024; Milionis et al., 2024; Lehar and

Parlour, 2025). In our setting, by contrast, issuer-sponsored depth crowds in external LPs only to a limited extent. Low turnover constrains fee revenue, and issuer-provided yield subsidies on deposited RWA tokens generate liquidity that does not scale with market activity. External LP participation is therefore tilted toward harvesting issuer-subsidized income rather than supplying depth for market making.

The constraining factor is scale: even if all trades in our data were theoretically routed through AMMs at the observed 0.25% fee, aggregate LP revenue would remain modest, approximately \$103k in total or about \$45 per active token-year, which at a typical price of \$50 and for balanced buy and sell orders translates into $\gamma = 0.9$.⁶ Formula (13) illustrates the consequences. For realistic and observable capital costs and RWA yields of $r = 0.09$ and $\mu = 0.09$, the maximum profitable liquidity provision L' is 12 (resp. 20) tokens for issuer-provided liquidity of 5 tokens ($L_R = 5$) (resp. 15). In other words, as liquidity providers tend to stay far below these values, the traded volume so far is insufficient to support persistent depth in the absence of additional incentives.

As emphasized by Park and Stinner, 2025, competition for liquidity in DeFi is intense. In tokenized RWA markets, permissioned participation can further limit liquidity mobility and shrink the set of eligible LPs. As a result, effective designs for sustaining AMM depth in inherently low-turnover, friction-constrained environments remain an important open challenge.

5.3. *Difference-in-differences analysis: fragmentation under stress*

AMM vs. P2P. The collapse of the TerraUSD (UST) stablecoin on May 7, 2022 provides a sharp, externally imposed shock to decentralized markets in DeFi.⁷ Importantly, 2022 is the only period in our sample in which AMM and P2P mechanisms are comparable in trading volume, while none of the tokens in our data are directly paired with UST or Luna, ruling out mechanical price transmission. Instead, the shock propagated through system-wide execution frictions, that are orthogonal to real-estate fundamentals of our observed RWA tokens.

We measure execution costs using fixed-size price impact (1 token or \$50), defined as the implied execution cost of trading a representative quantity of tokens under each mechanism. Price impact captures liquidity conditions implied by each mechanism’s pricing rule and available depth. By holding the trade size fixed, this measure isolates mechanism-driven liquidity costs rather than endogenous sorting in realized trades. To quantify the effect of the

TerraUSD collapse, we estimate the following difference-in-differences (DiD) specification:

$$Y_{i,t,m} = \beta_1 \text{AMM}_m + \beta_2 (\text{AMM}_m \times \text{Post}_t) + \mathbf{C}'_{i,t} \boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{i,t,m}, \quad (16)$$

where $Y_{i,t,m}$ denotes the 1-token price impact (in bps) for token i on day t under mechanism m . The indicator AMM_m equals one for AMM executions, and Post_t equals one for observations after May 7, 2022. Token fixed effects α_i absorb all time-invariant heterogeneity across assets, while day fixed effects δ_t absorb aggregate shocks common to all tokens. Standard errors are clustered at the token level. Control variables include rolling 7-day price volatility and volume-weighted average transaction fees to capture time-varying execution conditions. The coefficient of interest, β_2 , identifies the differential change in AMM execution costs relative to P2P trading following the Terra collapse for the same token.

[«Insert Table 6»]

Table 6 reports the DiD estimates for the 1-token price impact. Column (1) shows results for the full sample. Even prior to the shock, AMM execution is associated with substantially higher implied price impact than P2P trading, reflecting thinner liquidity. Following the Terra collapse, AMM execution costs increase sharply. The interaction coefficient $\text{AMM} \times \text{Post}$ indicates an additional increase of approximately 506 basis points in the implied cost of executing a 1-token trade on AMMs relative to P2P markets. Columns (2) and (3) split the sample by the median pre-event AMM pool size. The post-shock increase in AMM price impact is concentrated in small pools. For pools below the median size, the implied execution cost rises by more than 2,280 basis points, whereas for large pools the increase is economically much smaller, though still statistically significant. This monotonic attenuation across pool size directly implicates liquidity depth as the key state variable governing AMM resilience under stress. Columns (4) and (5) report analogous splits by underlying total asset size (USD). The post-Terra collapse increase in AMM execution costs is substantially larger for small assets than for large assets, consistent with lower turnover and thinner effective liquidity for smaller and more heterogeneous RWAs.

Figure 8 presents event-study estimates for the 1-token price impact. When aggregating across all pools, average effects evolve more smoothly around the Terra collapse. In contrast, event studies restricted to small pools (below median) show a sharp and persistent upward

shift in execution costs around the shock date. Event-study estimates show no evidence of differential pre-trends in execution costs across mechanisms, supporting the parallel-trends assumption.

[«Insert Figure 8»]

P2P vs. Buyback. On July 22, 2025, a Detroit court issued a Temporary Restraining Order (TRO) requiring rental collections for a subset of underlying properties to be redirected into escrow accounts until certificates of compliance were obtained.⁸ As a direct consequence, the issuer suspended the Buyback for all affected tokens. This legal intervention provides a plausibly exogenous shock to the issuer’s redemption facility, allowing us to examine how the withdrawal of this backstop facility affects liquidity conditions in P2P markets, which at that time were the dominant market mechanism. We estimate DiD regressions comparing P2P bid–ask spreads for tokens affected by the TRO to unaffected tokens before and after the event, with the following equation:

$$\begin{aligned} \text{Spread}_{it} = & \beta (\text{TRO affected}_i \times \text{Post}_t) \\ & + \gamma_1 \text{Volatility}_{it} + \gamma_2 \text{Fees}_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \end{aligned} \quad (17)$$

where Spread_{it} denotes the midpoint-normalized P2P bid–ask spread (in bps) for token i on day t . The indicator TRO affected_i equals one for tokens whose Buyback were suspended due to the TRO, and Post_t equals one for observations on or after July 22, 2025. Token fixed effects α_i absorb all time-invariant heterogeneity across assets, including location, property characteristics, and baseline liquidity, while day fixed effects δ_t capture aggregate shocks common to all tokens. Standard errors are clustered at the token level. Control variables include rolling 7-day price volatility and volume-weighted average transaction fees to account for time-varying market conditions.

Table 7 reports the results. While on average, affected tokens exhibit no significant increase in P2P bid–ask spreads following the TRO, the effect reveals when splitting the sample by asset size (USD). Spreads for large assets (above median in USD) widen substantially after their Buyback is suspended. In contrast, spreads for small assets (below median in USD) decrease, consistent with a reallocation of trading activity into P2P markets once their Buyback is suspended, increasing quote competition conditional on two-sided liquidity. For smaller asset sizes, the Buyback crowds out P2P trading. This effect is concentrated on the bid side: the share of tokens with standing P2P bids increases only among small assets, while it remains

essentially unchanged for large assets. The size-dependent effects are robust to controlling for a July 2, 2025 public disclosure listing the properties lacking certificates of compliance that were later subject to the TRO and issuer Buyback suspension.⁹

[«Insert Table 7»]

Figure 9 presents event-study estimates of P2P bid–ask spreads around the TRO. The estimates show no evidence of differential pre-trends. Following the TRO, bid–ask spreads for large assets increase persistently, indicating a sustained deterioration in liquidity. Spreads for small assets remain stable or decline modestly. The aggregate response masks these opposing effects across asset size groups.

[«Insert Figure 9»]

Overall, the evidence supports Hypothesis 3 and shows that fragmentation amplifies trading frictions through interacting mechanism-specific channels. Under systemic stress, AMMs continue to clear trades mechanically, but rising price impact reduces trading incentives and fee revenue, weakening external liquidity provision and accelerating migration toward information-sensitive outside options. P2P markets, while slower to execute, absorb this flow and exhibit greater price stability under stress. The issuer-provided Buyback interacts asymmetrically with these dynamics. It anchors prices and mitigates dispersion for large assets, but crowds out bilateral trading for small assets in normal times. When the Buyback is suspended, P2P liquidity deteriorates for large tokens as centralized support is withdrawn, while participation and quote competition increase for small tokens as bilateral trading becomes the marginal liquidity source. Liquidity outcomes under stress therefore depend on asset scale and the interaction between execution mechanisms.

6. Conclusion

This paper studies liquidity formation in a tokenized market for illiquid RWAs using a uniquely transparent dataset of 777,038 secondary-market transactions. This fully on-chain environment is the only setting in which three fundamentally different liquidity mechanisms coexist under a uniform token design. The central implication is structural: fragmentation across heterogeneous mechanisms does not merely redistribute order flow, but shapes how liquidity is financed and sustained. When turnover is low or market conditions deteriorate, deterministic pricing remains operational but becomes fragile as execution costs rise and fee-based liquidity provision contracts, shifting activity toward alternative mechanisms in a manner that depends on asset scale and issuer backstop liquidity. Fragmentation therefore places sharp limits on when deterministic market design can sustain depth and becomes an amplifier of trading frictions rather than a neutral allocation of liquidity.

While tokenization can facilitate trading, the question of liquidity creation remains unresolved and highlights trade-offs in the secondary market design of inherently illiquid but tokenized RWAs. When platforms rely on AMMs to provide instant, two-sided liquidity, viability becomes highly scale-dependent. For small, slow-moving, and heterogeneous RWAs, thin pools generate high price impact, making rule-based pricing fragile and liquidity provision unlucrative. In addition, whitelisting, and especially token-level whitelisting, fragments participation and limits trading intensity and liquidity provision, further constraining sustainable liquidity. Buyback acts as a capped, appraisal-based liquidity backstop that absorbs larger redemptions at the cost of market-based price formation. P2P trading dominates economic volume due to lower effective price impact, but resembles classical OTC search markets with uncertain execution and largely hidden buy-side liquidity. This leaves unresolved who bears the cost of visible, committed liquidity in fragmented tokenized markets for traditionally illiquid RWAs.

In conclusion, tokenization alone does not ensure liquid secondary markets for illiquid RWAs. When heterogeneous execution mechanisms coexist, liquidity formation is constrained by market design and incentive compatibility, leaving liquidity fragmented, fragile, and conditional on issuer support rather than fully endogenous. P2P sustains liquidity but resembles a digitized, technologically facilitated search market that does not guarantee continuous liquidity. Without improved mechanism design or deliberate choices about which frictions to tolerate, tokenized markets risk reproducing rather than resolving the liquidity constraints they seek to eliminate.

Notes

¹Note that price dispersion across mechanisms is captured by an additional parameter for transaction costs referring to the AMM price.

²Note that we do not use the usual “impermanent loss” concept, as its standard benchmark is holding the same quantities of the base and quote tokens outside of an AMM (i.e., a passive two-asset hold). This behavior may be typical of a crypto investor with a diversified token portfolio and is useful as a relative performance measure of providing liquidity versus holding that 50/50 portfolio. In our setting, however, the economically relevant decision is whether to commit fresh capital to acquire and tie up the inventories required to provide liquidity, or instead to hold the RWA token (or invest funds elsewhere). We therefore framework liquidity provision as an absolute profit problem and explicitly incorporate the opportunity/financing cost of capital at rate r . This automatically limits the liquidity that can be provided due to capital costs; in an approach based on impermanent loss, however, there are no such limits.

³This condition requires that the investor’s capital costs exceed the AMM’s passive return. This is empirically plausible in real-world settings, as otherwise a form of unbounded arbitrage would emerge (raising capital at rate r and earning $\frac{\mu}{2}$).

⁴The total token count is 3,150,484. RealT reports retaining 5% of tokens at primary issuance (157,524 retained tokens), leaving approximately 3 million tokens in free float.

⁵A notable exception occurs in February 2024, when the issuer withdrew most of its per-pool contribution, reducing the seed position from 20 to 4 RWA tokens (approximately \$1,600 at mean token prices).

⁶See Equation 12. We imply the total USD trading volume of \$41.2M (see Table 1). Active token-years are token-years with at least one trade (2,258). USD per active token-year are computed as total theoretical fee revenue divided by 2,258.

⁷At the time, UST was the third-largest stablecoin with a market capitalization of roughly \$18 billion. The collapse triggered ecosystem-wide volatility, liquidity withdrawals, fee spikes, and pricing distortions, including spillovers to other major stablecoins (Uhlig, 2022; Lee et al., 2023).

⁸For more details on the lawsuit, see <https://detroitmi.gov/news/real-token-properties-are-subject-sweeping-lawsuit>.

⁹The results are available in the replication package. For the July 2 filing/announcement document, see <https://outliermedia.org/wp-content/uploads/2025/07/City-v.-Real-Token-Complaint-1-1.pdf>.

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Tables

Table 1: Trade statistics by market mechanism

Statistic	N	Mean	St. Dev.	Min	10%	Median	90%	Max
<i>Panel A: AMM</i>								
Trade size (RWA token)	555,169	0.099	0.668	0.000	0.009	0.016	0.100	300.000
Trade volume (USD)	555,169	6.261	43.915	0.000	0.488	0.824	6.387	3,782
Total fees (USD)	555,169	0.048	0.755	0.000	0.002	0.003	0.022	117.244
Trade price (USD/RWA token)	555,169	54.551	15.752	0.184	47.189	52.835	61.287	2,394
<i>Panel B: P2P</i>								
Trade size (RWA token)	207,885	3.068	14.652	0.000	0.039	1.000	6.000	1,771
Trade volume (USD)	207,885	166.870	788.185	0.000	2.000	51.610	333.960	89,970
Total fees (USD)	207,885	0.192	2.827	0.000	0.000	0.000	0.003	148.697
Trade price (USD/RWA token)	207,885	56.422	16.444	1.000	49.000	52.555	65.000	617.000
<i>Panel C: Buyback</i>								
Trade size (RWA token)	13,984	4.134	6.431	0.000	0.157	2.000	10.000	40.000
Trade volume (USD)	13,984	219.434	341.807	0.000	8.041	100.600	520.140	2,000
Total fees (USD)	13,984	7.690	11.478	0.000	0.256	3.057	19.218	118.082
Trade price (USD/RWA token)	13,984	53.189	13.000	44.190	49.560	50.715	54.950	185.800
<i>Panel D: Total Market</i>								
Trade size (RWA token)	777,038	0.966	7.771	0.000	0.009	0.029	2.000	1,771
Trade volume (USD)	777,038	53.066	418.581	0.000	0.491	1.305	100.000	89,970
Total fees (USD)	777,038	0.224	2.437	0.000	0.000	0.003	0.021	148.697
Trade price (USD/RWA token)	777,038	55.027	15.918	0.184	47.818	52.728	62.437	2,394
<i>Cumulative trading volumes (USD, full sample)</i>								
Total trading volume (AMM, USD)								3,476,186
Total trading volume (P2P, USD)								34,689,710
Total trading volume (Buyback, USD)								3,068,572
Total trading volume (All, USD)								41,234,467

Note: Panels A–C report summary statistics for *Trade size (RWA token)*, *Trade volume (USD)*, *Total fees (USD)*, and *Trade price (USD/RWA token)* by mechanism (AMM, P2P, Buyback). Panel D reports the same variables for the full sample and additionally the *Cumulative trading volumes (USD)* by mechanism. We report the 10th percentile and 90th percentile to highlight distributional skewness. All statistics exclude missing observations.

Table 2: Descriptive statistics

Statistic	N	Mean	St. Dev.	Min	10%	Median	90%	Max
Trades-intraday perspective								
Trade size (RWA token)	777,038	0.97	7.77	0.00	0.01	0.03	2.00	1,771
Trade volume (USD)	777,038	53.07	418.58	0.00	0.49	1.30	100.00	89,970
RWA token buy	777,038	0.38	0.49	0.00	0.00	0.00	1.00	1.00
Ethereum blockchain	777,038	0.01	0.10	0.00	0.00	0.00	0.00	1.00
Price dispersion-intraday perspective								
Price signal AMM (USD)	776,532	62.44	5,014	0.18	47.77	52.67	62.99	4,417,601
Price signal P2P bid offer (USD)	167,381	51.39	13.46	29.84	44.76	49.64	55.33	218.20
Price signal P2P ask offer (USD)	635,475	56.66	11.19	35.00	50.57	55.00	62.80	241.27
Price signal Buyback (USD)	777,038	51.14	13.99	0.00	49.12	50.79	54.42	185.80
AMM liquidity-intraday perspective								
AMM pool liquidity bid (USD)	776,532	1,144	2,913	0.00	47.36	847.14	1,577	89,069
AMM pool liquidity ask (USD)	776,532	1,144	2,913	0.00	47.36	847.14	1,577	89,069
Total AMM pool liquidity (USD)	776,532	2,287	5,826	0.00	94.71	1,694	3,155	178,139
AMM slippage (%)	555,169	-0.012	0.023	-1.000	-0.034	-0.005	-0.003	0.000
LP trading fee revenue (USD)	555,169	0.019	0.132	0.000	0.001	0.002	0.019	11.346
P2P liquidity-intraday perspective								
P2P bid offer depth (USD)	167,381	1,352	1,836	0.00	50.05	517.47	3,895	17,525
P2P ask offer depth (USD)	635,475	6,005	16,426	0.00	57.00	992.60	14,336	433,436
Total P2P offer depth (USD)	646,415	6,253	16,431	0.00	58.90	1,184	14,717	433,686
Time on market since offer creation (days)	207,885	17.89	48.40	0.00	0.00	1.81	48.16	1,140
Stress event variables-intraday perspective (± 100 -days around the shock)								
1-token price impact (bps)	235,404	1,640	2,501	-10,000	256.11	521.33	7,655	9,101
Bid-ask spread (bps)	64,912	1,472	730.97	0.00	605.60	1,393	2,438	4,919
Asset size (USD)	752	215,595	327,988	48,080	62,440	84,712	590,675	5,000,000
TRO affected	752	0.73	0.45	0.00	0.00	1.00	1.00	1.00
Market control variables at trade-daily perspective								
RWA token price volatility (7-day)	185,940	0.08	0.16	0.00	0.02	0.05	0.15	3.18
Market-to-appraisal ratio	188,193	1.06	0.05	0.93	0.99	1.06	1.11	1.19
Cumulative return ETH (one week before)	188,193	0.01	0.11	-0.52	-0.11	0.00	0.15	0.67
Average ETH tx fee (USD)	188,193	8.36	12.20	0.07	0.61	4.00	23.57	200.06
1m Treasury yield Δ	188,193	0.00	0.05	-0.55	-0.02	0.00	0.03	1.06
10y Treasury yield Δ	188,193	0.00	0.06	-0.30	-0.08	0.00	0.08	0.29
ADS Index Δ	188,193	-0.10	0.96	-26.69	-0.47	-0.12	0.36	9.58
S&P Case Shiller Index Δ	188,193	0.01	0.01	-0.01	-0.00	0.00	0.02	0.03

Note: This table reports the number of observations, mean, standard deviation, minimum, 10th percentile, median, 90th percentile, and maximum for the intradaily (transaction-level) and daily aggregated perspectives. All variables and the corresponding calculation are defined in Appendix Table A.5.

Table 3: Transaction-level mechanism choice

	AMM trade (1)	P2P trade (2)	Buyback trade (3)	AMM trade (4)	P2P trade (5)	Buyback trade (6)
log(Trade volume, USD)	-0.0631*** (0.0025)	0.0555*** (0.0025)	0.0076*** (0.0005)	-0.0606*** (0.0017)	0.0533*** (0.0016)	0.0090*** (0.0005)
Blockchain fee (USD)	-0.0079*** (0.0023)	0.0008 (0.0014)	0.0070*** (0.0016)	-0.0038*** (0.0005)	-0.0002 (0.0004)	0.0032*** (0.0005)
Market-to-appraisal ratio	0.1539*** (0.0554)	-0.1382** (0.0542)	-0.0158** (0.0077)	-0.1584*** (0.0607)	0.3180*** (0.0775)	0.0264* (0.0150)
Average ETH tx fee (USD)	0.0002*** (0.0001)	-0.0002*** (0.0001)	0.0000 (0.0000)	0.0006*** (0.0001)	-0.0007*** (0.0001)	0.0000 (0.0000)
AMM slippage (wavg)	-0.7786*** (0.0660)	0.6969*** (0.0607)	0.0817*** (0.0075)			
RWA token price volatility (7-day)	-0.0203 (0.0180)	0.0151 (0.0172)	0.0053*** (0.0020)	0.0095 (0.0139)	-0.0069 (0.0149)	0.0114*** (0.0043)
Cumulative return ETH (1 week)	-0.0610*** (0.0097)	0.0594*** (0.0097)	0.0016 (0.0013)	-0.0107 (0.0100)	-0.0051 (0.0117)	-0.0041 (0.0027)
1m Treasury yield Δ	-0.0034 (0.0260)	-0.0036 (0.0268)	0.0071* (0.0037)	-0.1095*** (0.0220)	0.0966*** (0.0233)	0.0120 (0.0076)
10y Treasury yield Δ	-0.0256** (0.0126)	0.0244* (0.0127)	0.0012 (0.0015)	-0.0443*** (0.0132)	0.0345** (0.0144)	0.0066* (0.0036)
ADS index Δ	0.0206 (0.0165)	-0.0126 (0.0158)	-0.0080** (0.0031)	-0.0096 (0.0213)	-0.0168 (0.1257)	0.0413 (0.0281)
AMM pool liquidity bid (kUSD)				0.0034*** (0.0006)	-0.0034*** (0.0010)	-0.0004** (0.0002)
P2P ask offer depth (kUSD)					0.0005*** (0.0001)	0.0000 (0.0000)
Time on market since offer creation (days)					0.0029*** (0.0001)	
Token FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	603,527	603,527	603,527	764,843	628,229	628,229
R ²	0.355	0.338	0.044	0.643	0.634	0.076
Within R ²	0.174	0.144	0.034	0.174	0.195	0.023

Note: This table reports transaction-level regressions of mechanism choice on trade characteristics, market conditions, and mechanism-specific liquidity measures. Cluster-robust standard errors (token level) are reported in parentheses. All regressions are estimated at the transaction level using linear probability models with token and year-month fixed effects. The main explanatory variable is the logarithm of the trade's USD notional value. Columns 1–3 report baseline specifications including log(Trade volume), blockchain fees, AMM slippage, market-to-appraisal ratios, ETH fees, rolling 7-day price volatility, ETH returns, Treasury yield changes, and ADS index changes. Columns 4–6 replace AMM slippage with mechanism-specific liquidity (scaled in thousand USD): AMM bid-side pool liquidity for AMM routing, P2P ask-side offer depth and time on market of the active listing for P2P routing, and both AMM bid-side pool liquidity and P2P ask offer depth. Note that AMM bid and ask side is symmetric given the constant-product AMM mechanism. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 4: Liquidity provision across market mechanisms

Statistic	N	Mean	St. Dev.	Min	10%	Median	90%	Max
<i>Panel A: AMM liquidity provision (event-level)</i>								
<i>Issuer LP</i>								
Deposit (USD)	954	1,181	1,536	1.56	101.48	207.76	2,075	18,939
Withdrawal (USD)	778	1,483	6,301	0.01	2.47	412.71	1,671	82,684
<i>External LPs</i>								
Deposit (USD)	4,597	225.50	1,233	0.00	3.08	85.11	456.98	49,383
Withdrawal (USD)	3,321	435.16	2,974	0.00	10.01	102.89	562.74	136,684
<i>Total liquidity activity</i>								
Deposit (USD)	5,551	389.67	1,339	0.00	4.58	101.12	1,275	49,383
Withdrawal (USD)	4,099	634.09	3,855	0.00	9.96	109.23	1,582	136,684
Protocol-internal activity (USD)	3,079	159.61	1,049	0.00	1.99	54.66	309.88	50,783
<i>Panel B: AMM liquidity provision (pool-week panel)</i>								
External liquidity (USD)	23,833	494.76	3,957	0.00	0.36	104.55	398.24	131,308
Issuer liquidity (USD)	23,833	1,580	3,292	0.00	40.02	1,259	2,423	73,341
LP trading fee revenue (USD)	23,833	0.38	2.23	0.00	0.01	0.08	0.49	141.50
AMM token exchange rate volatility	23,833	0.049	0.091	0.000	0.007	0.030	0.089	4.399
<i>Panel C: P2P two-sided quote provision (wallet-token pair level)</i>								
Bid and ask offer overlap (days)	835	23.17	39.60	0.00	1.18	9.01	67.61	651.96
Price signal bid offer (USD)	835	50.56	14.66	33.22	45.10	49.02	52.55	215.00
Price signal ask offer (USD)	835	56.11	17.22	40.74	50.18	52.65	61.48	228.99
Spread (simple, %)	835	11.04	8.28	-2.80	2.94	9.69	22.15	57.74
P2P bid offer depth (USD)	835	382.01	751.48	0.00	96.77	199.09	857.53	10,526
P2P ask offer depth (USD)	835	601.81	3,537	0.01	20.14	111.87	794.79	76,987
Total offer depth (USD)	835	185.54	403.54	0.00	14.26	100.10	300.30	6,217

Note: Panel A reports event-level AMM liquidity provision across all pools, split into *Issuer LP*, *External LPs*, and *Total*. Deposit and withdrawal amounts are measured as two-sided USD flows, computed as twice the USD change in the pool's quote-token reserve under the constant-product design. Activity initiated by the protocol's designated fee-collector (LevinMaker) wallet is reported separately. Panel B reports pool-week summary statistics computed on the weekly regression sample (restricted to pool-weeks with at least two swaps in week t and defined within-week volatility). Panel C summarizes *P2P two-sided quote provision* by wallet-token. Wallets qualify if realistic bid and ask offers overlap at least once and both sides have positive posted depth in at least one overlap window. Overlap is total bid-ask overlap duration (days). Bid/ask price signals are liquidity-weighted time-averaged posted prices (USD) over overlap windows; the spread is the percentage difference between the ask and bid price signals and can be negative if the bid exceeds the ask. Total offer depth is the smaller of the bid- and ask-side posted depth.

Table 5: External liquidity provision in AMM pools

	log(External liquidity _{t+1} , USD)	
	(1)	(2)
log(LP fee revenue _t , USD)	1.097*** (0.165)	0.656*** (0.119)
log(Issuer liquidity _t , USD)	0.683*** (0.034)	0.873*** (0.031)
log(Issuer liquidity _t , USD) × Low-trading-fee env.	0.094* (0.053)	0.166*** (0.037)
AMM token exchange rate volatility _t	-2.382*** (0.266)	-1.469*** (0.213)
Pool FE	No	Yes
Year-Month FE	No	Yes
Observations	23,833	23,833
R ²	0.299	0.371

Notes: This table reports pool-week regressions of $\log(\text{External liquidity}_{t+1})$, computed from end-of-week external liquidity provision in the subsequent calendar week. Trading fee revenue is the weekly sum of LP fees, and issuer liquidity is end-of-week issuer-provided liquidity. Volatility is the standard deviation of the post-swap log spot exchange rate (quote per base) across swap events within the pool-week. The low-trading-fee environment is proxied by the negative, mean-centered log trading-fee revenue (higher values = lower-fee weeks). Log denotes the natural logarithm; coefficients on logged regressors are interpretable as elasticities. Column (1) is pooled OLS and reports overall R^2 ; column (2) includes pool and year-month fixed effects and reports within- R^2 . Standard errors are clustered by pool. *, **, *** denote significance at the 10%, 5%, and 1% levels. Sample is restricted to pool-weeks with at least two swaps. Results are robust to alternative minimum swap-count thresholds per pool-week used to compute volatility and to estimation on a daily panel.

Table 6: 1-token price impact around the Terra collapse (AMM vs. P2P)

	1-token price impact (bps)				
	All tokens	Pool size split (USD)		Asset size split (USD)	
		Small pools (< median)	Large pools (> median)	Small assets (< median)	Large assets (> median)
	(1)	(2)	(3)	(4)	(5)
AMM	1,045.3*** (112.11)	2,598.6*** (368.87)	571.0*** (23.34)	2,618.2*** (319.63)	756.1*** (66.59)
RWA token price volatility (7-day)	73.0 (254.38)	400.9** (182.58)	-20.7 (53.19)	696.7** (314.95)	100.5 (286.41)
Total fees (USD, wavg)	2.1 (2.99)	0.3 (2.28)	0.2 (0.31)	-9.7 (6.01)	3.3 (2.07)
AMM $\times$ Post	506.0*** (152.63)	2,284.7*** (356.30)	82.7** (39.09)	598.1** (295.16)	216.8** (104.91)
Token FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	235,404	71,506	140,982	78,745	156,659
R^2	0.085	0.353	0.436	0.154	0.128

Notes. This table reports DiD estimates for the 1-token price impact (bps) comparing execution costs on AMMs relative to P2P markets around the Terra collapse on May 7, 2022. Column (1) reports results for the full sample. Columns (2) and (3) split the sample by the median pre-event AMM pool size (USD). Columns (4) and (5) split the sample by the median underlying asset size (USD). All specifications include token and day fixed effects. Standard errors are clustered at the token level. ***, **, and * denote significance at the 1%, 5%, and 10% levels. We report within- R^2 . The results remain robust when all fixed-size price impact measures are winsorized at the 1st and 99th percentiles to limit the influence of extreme observation and the corresponding estimates are available in the replication package.

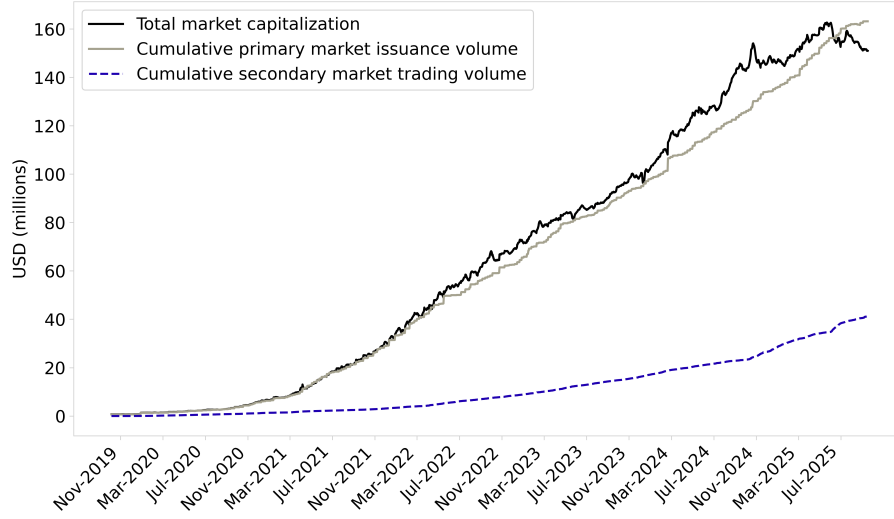
Table 7: P2P Bid–ask spreads around the Temporary Restraining Order (P2P vs. Buyback)

	Bid–ask spread (bps)		
	All tokens	Asset size split (USD)	
		Small assets (< median)	Large assets (> median)
	(1)	(2)	(3)
RWA token price volatility (7-day)	-39.30 (60.60)	54.12 (54.22)	-78.63 (83.00)
Total fees (USD, wavg)	-10.48*** (2.103)	-10.24*** (1.629)	-10.48*** (2.935)
TRO affected × Post	191.4 (117.2)	-193.9*** (61.99)	271.1** (123.5)
Token FE	Yes	Yes	Yes
Day FE	Yes	Yes	Yes
Observations	64,694	19,203	45,491
R^2	0.013	0.009	0.023

Notes: This table reports DiD estimates of P2P bid–ask spreads around the Temporary Restraining Order (TRO) issued on July 22, 2025, which led to the issuer-initiated suspension of the Buyback mechanism for 546 RWA tokens out of 752. The dependent variable is the midpoint-normalized bid–ask spread expressed in basis points. Treated tokens (*TRO affected*) are those for which Buyback was suspended due to the TRO. Column (1) reports results for the full sample; columns (2) and (3) split the sample by the median underlying asset size (USD). All specifications include token and day fixed effects and control for rolling 7-day price volatility and volume-weighted average transaction fees. Standard errors are clustered at the token level. ***, **, and * denote significance at the 1%, 5%, and 10% levels. We report within- R^2 . The results remain robust when all fixed-size price impact measures are winsorized at the 1st and 99th percentiles to limit the influence of extreme observations and the corresponding estimates are available in the replication package.

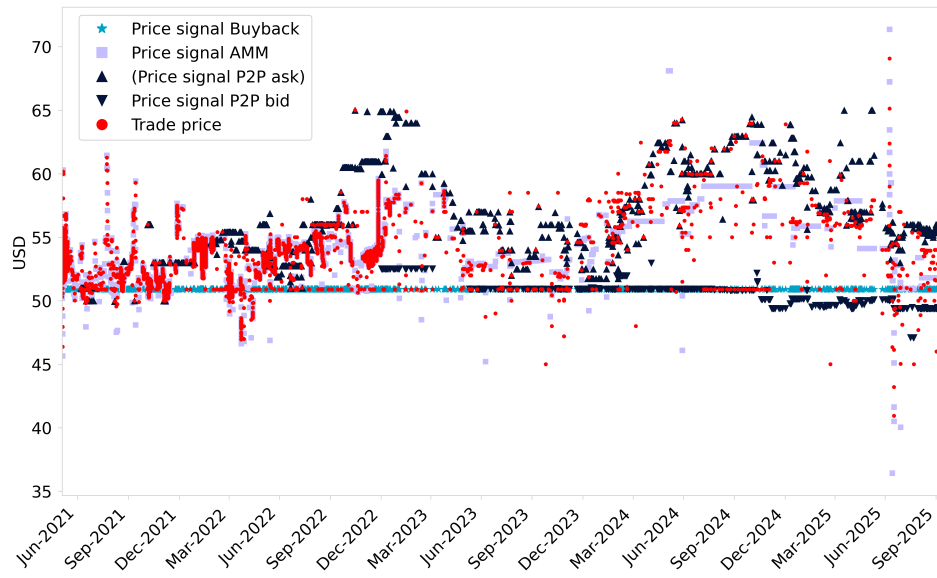
Figures

Figure 1: RealT's market evolution



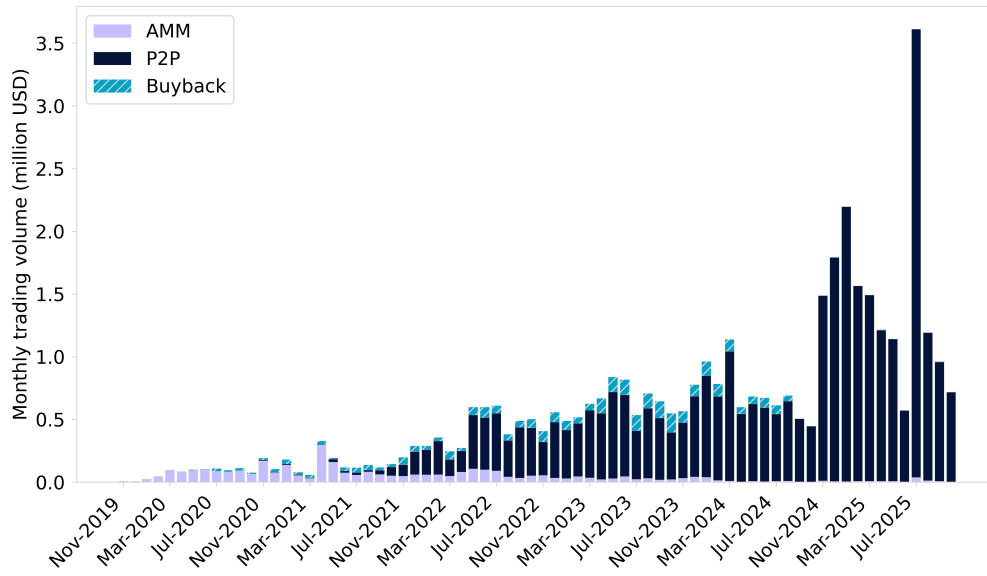
Note: This figure plots three global series for the RealT token universe (million USD). *Total market capitalization* is computed each day as the sum across tokens of total token supply times a 7-day moving average of the token's daily secondary-market volume-weighted average trading prices. *Cumulative primary market issuance volume* is the sum across tokens of the issuance value at STO. *Cumulative secondary market trading volume* is the cumulative sum of secondary-market trading volume. Values are forward-filled over time for missing observation days.

Figure 2: Price signals and realized trade prices for one tokenized RWA (sell-side)



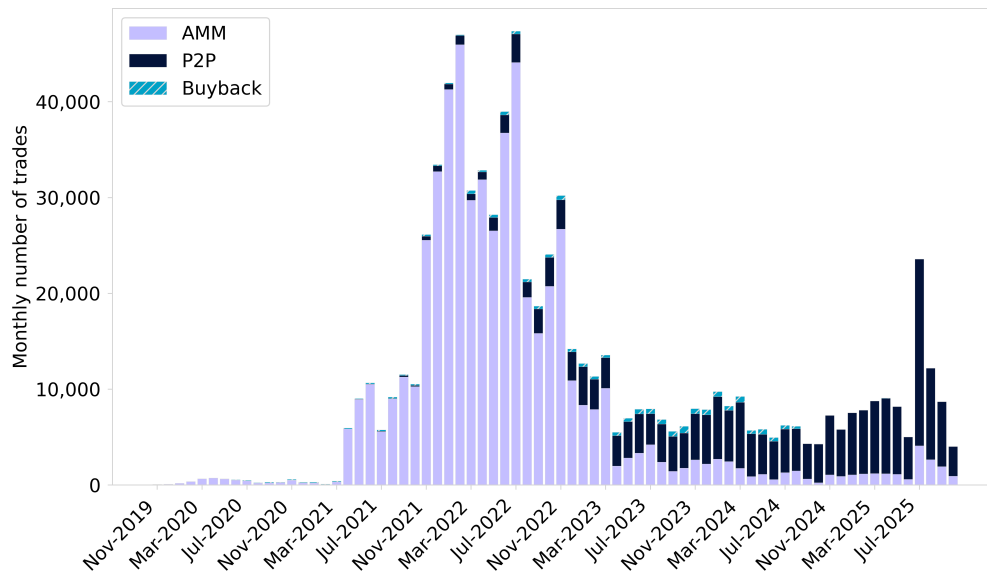
Note: This figure plots daily USD price signals and realized trade prices for one exemplary tokenized RWA on the *sell-side* in our sample. Observation counts (in parentheses) report the number of available observations underlying each series and sum to 20,497 price-signal observations and 7,024 realized trades (trade prices). The Buyback quotes based on appraisal values (a flat line over the sample period because no re-appraisal occurred; $N=7,024$), AMM quotes implied by pool reserves (identical for bid and ask side; $N=7,024$), P2P bid ($N=968$) and ask ($N=5,481$) signals computed as daily median offer prices from reconstructed offer books, and realized trade prices ($N=7,024$). Parentheses denote P2P ask signals, which are not executable on the sell-side.

Figure 3: Trading volume by market mechanism



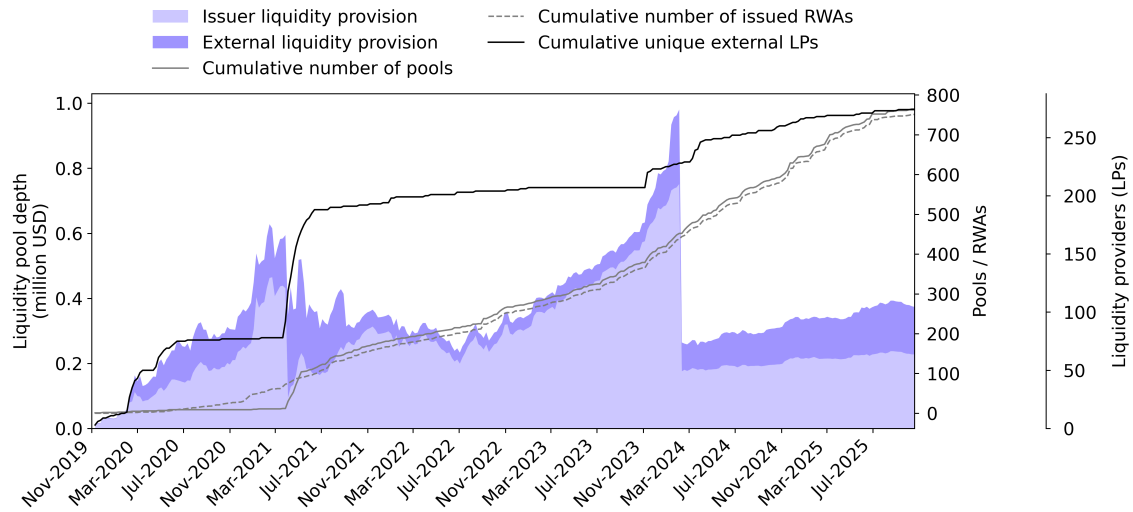
Note: This figure presents the monthly trading volume in million USD categorized by market mechanisms.

Figure 4: Number of trades by market mechanism



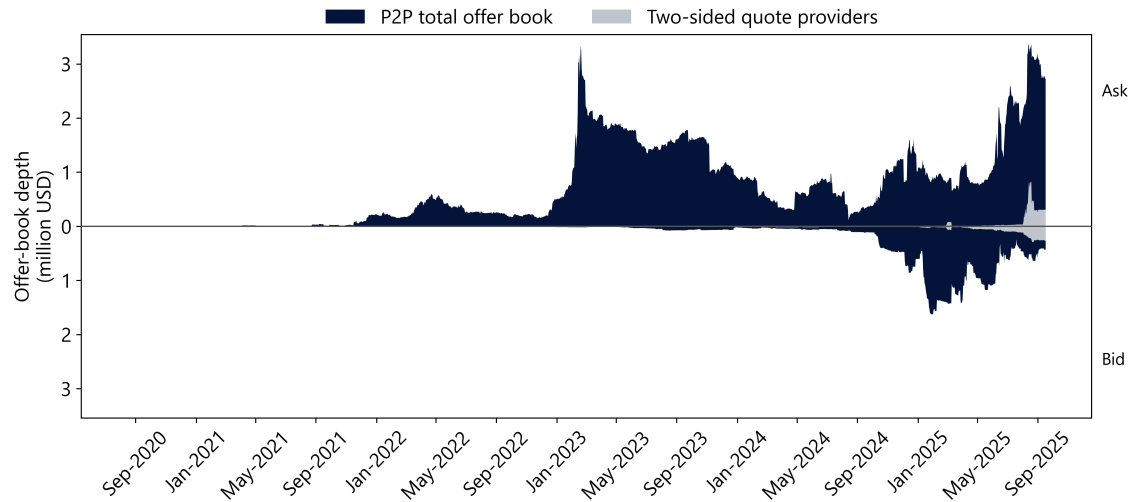
Note: This figure presents the monthly number of trades categorized by market mechanisms.

Figure 5: AMM liquidity: Issuer vs external providers



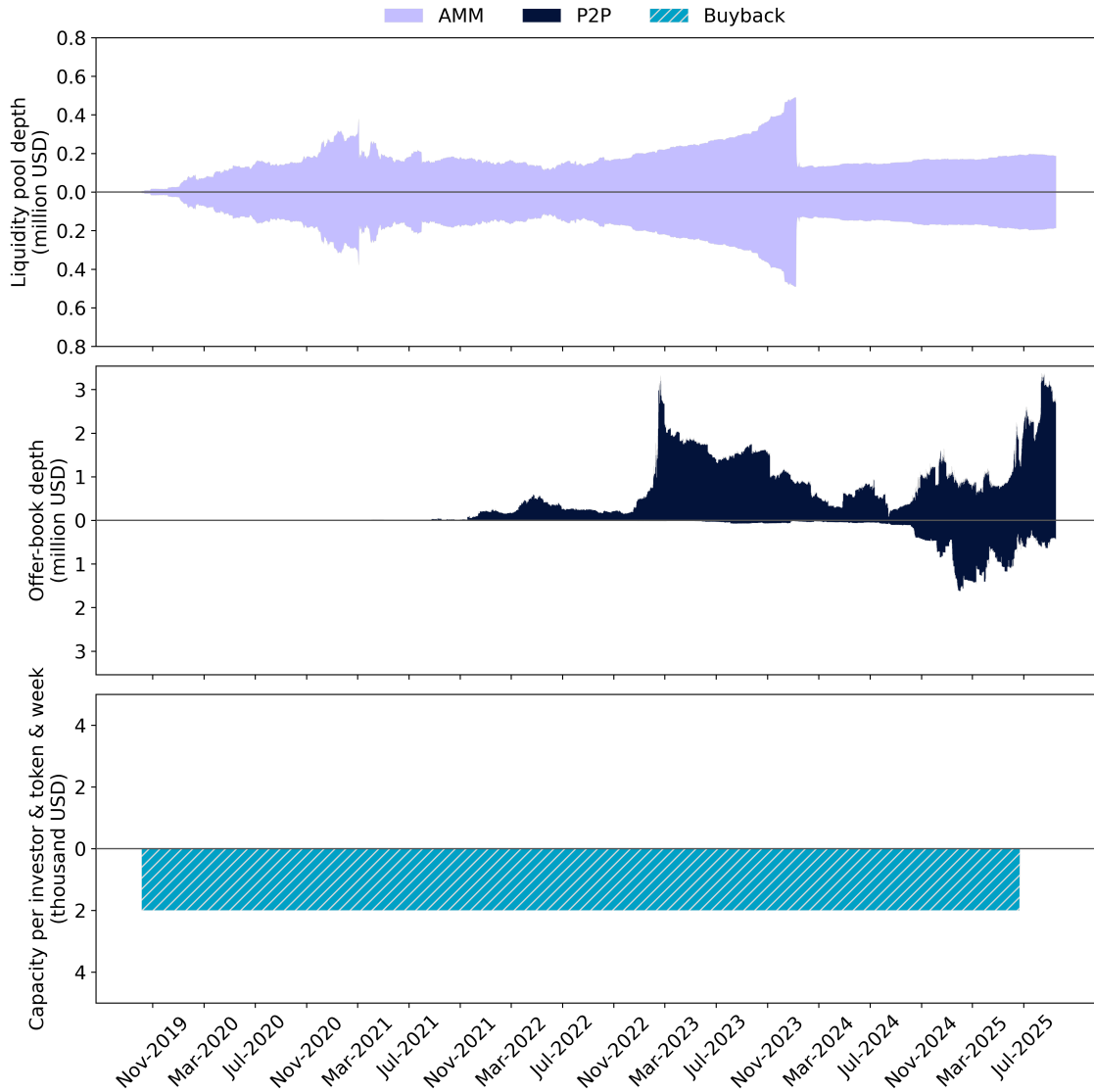
Note: This figure plots weekly total AMM depth decomposed into *issuer* and *external liquidity provision*. Since 2022, pools are paired exclusively with USDC (see Appendix A.2). The three major declines in AMM depth correspond to (i) the LEVIN/USD collapse in April 2021 (the then-dominant quote token), (ii) the ETH/USD drawdown during the Terra collapse in mid-2022 affecting (W)ETH-quoted pools, and (iii) RealT's regime shift and near-complete liquidity withdrawal across all pools in February 2024.

Figure 6: P2P liquidity: Two-sided quote providers



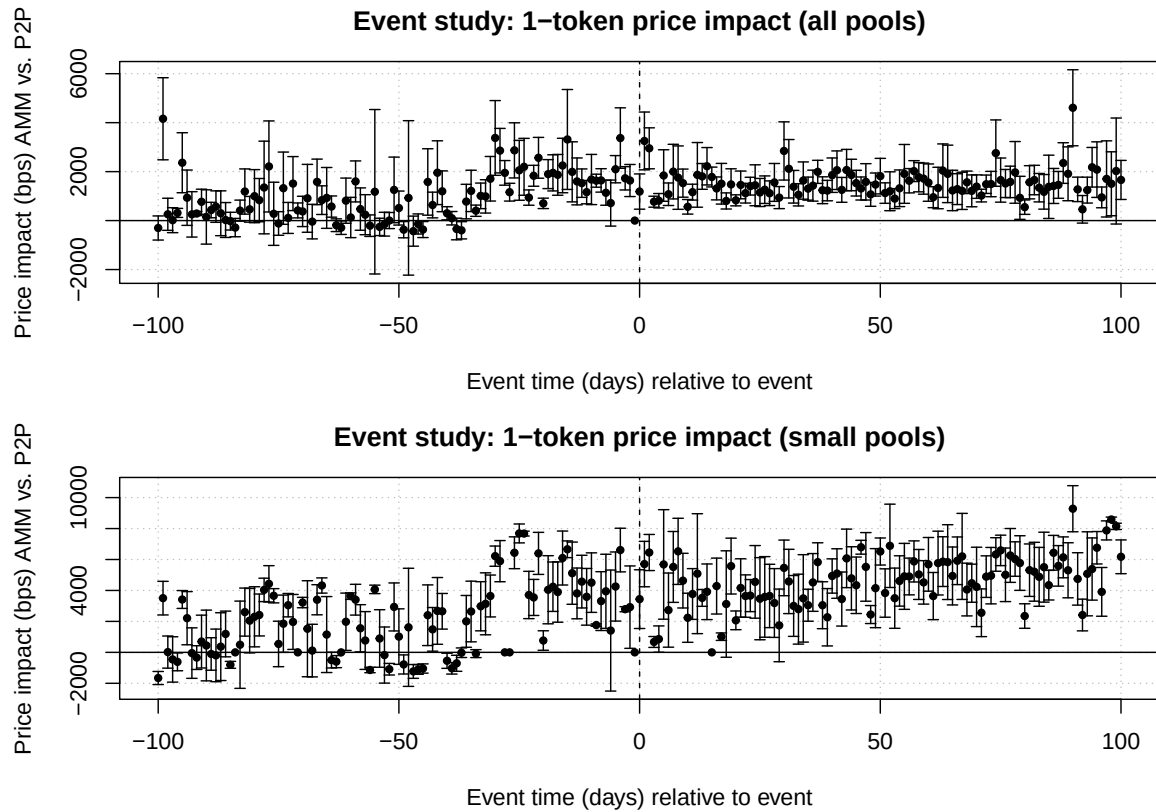
Note: This figure plots P2P liquidity measured as the total posted notional depth on the relevant side of the offer book (ask-side offers to sell and bid-side offers to buy), summed across all active listings on each day; unposted or hidden liquidity is not included. The lighter gray area isolates the portion of depth posted by wallets that simultaneously maintain bid and ask offers for the same token (*two-sided quote providers*)

Figure 7: AMM liquidity pool depth, P2P offer-book depth, and Buyback capacity



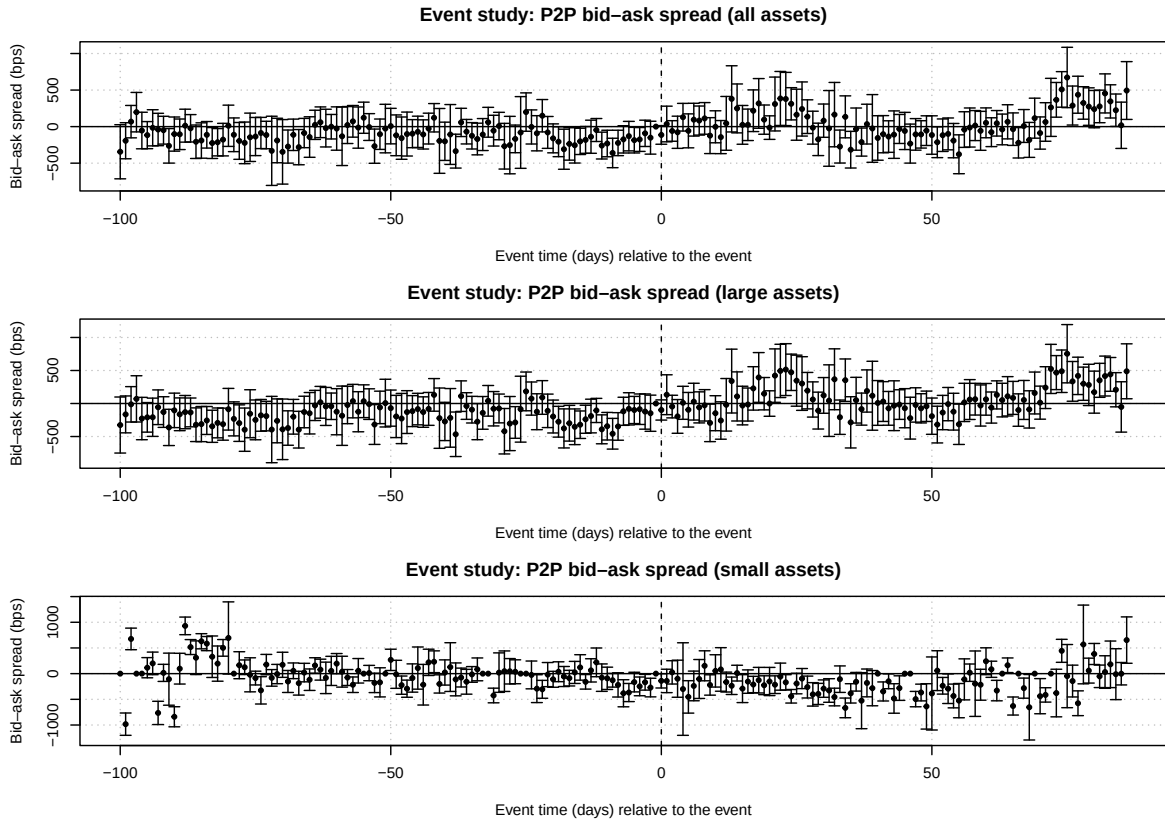
Note: This figure plots aggregate daily liquidity available across the three secondary-market mechanisms. AMM liquidity is measured as the total two-sided notional depth implied by constant-product pool reserves under pool-implied pricing. For each pool, we proxy two-sided depth as $2 \times$ the USD value of the quote-token reserve (valued at contemporaneous quote-token USD prices) and aggregate across all active pools each day. The total is mirrored at the x-axis for illustration purpose of ask and bid side. P2P liquidity is measured as the total posted notional depth on the relevant side of the offer book (ask-side offers to sell and bid-side offers to buy), summed across all active listings on each day; unposted or hidden liquidity is not included. Buyback capacity is derived from the issuer's deterministic redemption facility and is measured as the aggregate daily executable notional implied by platform-imposed quantity caps and prevailing appraisal-based Buyback prices. All series are constructed at the daily frequency and represent potential executable liquidity rather than realized trading volume. Light gray shaded areas mark periods in which only two mechanisms operate concurrently; in particular, the Buyback was suspended for most (though not all) tokens following the TRO.

Figure 8: Event study of 1-token price impact around the Terra collapse



Notes: This figure plots difference-in-differences event-study estimates of AMM price impact relative to P2P markets around the Terra collapse on May 7, 2022. The outcome variable is the implied price impact of executing a trade of one token (\$50), expressed in basis points. Coefficients are normalized to zero on day -1 and are estimated from regressions including token and day fixed effects, as well as controls for rolling 7-day price volatility and volume-weighted average transaction fees and 95% confidence intervals based on standard errors clustered at the token level. The top panel reports results for all pools, while the bottom panel restricts the sample to tokens associated with small AMM pools (below median pool size in USD).

Figure 9: Event study of P2P bid–ask spreads around the Temporary Restraining Order



Notes: This figure plots difference-in-differences event-study estimates of P2P bid–ask spreads around the Temporary Restraining Order (TRO) issued on July 22, 2025. The outcome variable is the midpoint-normalized P2P bid–ask spread, expressed in basis points. Coefficients are normalized to zero on day -1 and are estimated from regressions including token and day fixed effects, as well as controls for rolling 7-day price volatility and volume-weighted average transaction fees. Shaded areas represent 95% confidence intervals based on standard errors clustered at the token level. The top panel reports results for all assets, the middle panel for large assets (above median in USD), and the bottom panel for small assets (below median in USD), where asset size is defined based on the median underlying asset value in USD.

A.1. RWA market and liquidity mechanisms

This section positions RealT within the broader landscape of RWA tokenization platforms. RealT’s joint use of liquidity-pool-based AMMs, bilateral P2P, and issuer redemption-style Buyback mechanisms, combined with fully transparent on-chain record-keeping, provides a comprehensive setting to study liquidity formation under the fragmentation often encountered in DeFi. In particular, the RealT ecosystem constitutes a natural laboratory for tokenized illiquid RWAs, offering uniquely rich transaction-level data across coexisting secondary-market mechanisms under a uniform token design. Table [A.1](#) summarizes selected tokenized RWA platforms by underlying asset class and their overarching secondary-market liquidity mechanisms, with platform-specific implementations. Among the platforms summarized, RealT uniquely enables observation of all three prominent mechanism categories over an extended horizon, with the full history of trades and liquidity-relevant events recorded on public blockchains since 2019. Many other platforms rely on one or two of these same mechanism categories, making insights from the RealT setting naturally transferable.

Table A.1: Overview of RWA tokenization platforms and their liquidity mechanisms

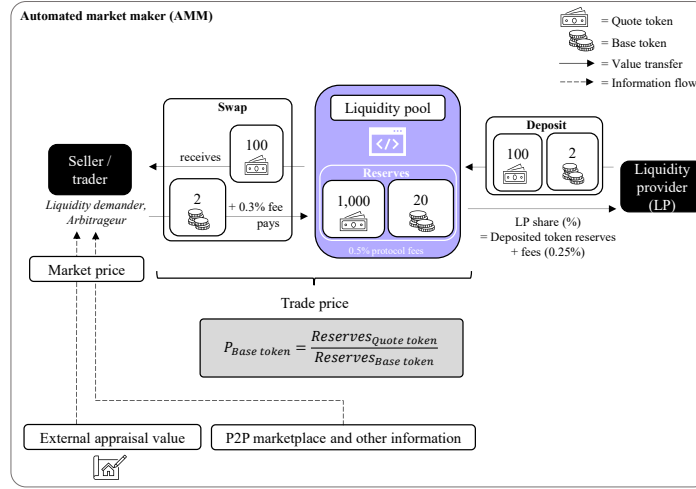
Platform	Asset class	Liquidity mechanism	Description
RealT	Real Estate	AMM/Pool-based (Uniswap, Levinswap), P2P (issuer-non-operated), Buyback/Redemption (issuer)	Tokenized real estate platform using multiple DeFi-native liquidity mechanisms. High transparency, core case study for this paper.
Lofty.ai	Real Estate	P2P (centralized marketplace listings) / AMMs/Pool-based (Proactive Market Maker)	Real estate tokenization platform where individual issuers can tokenize properties and sell shares on a centralized secondary market, with valuation-based pricing and pool-based liquidity and P2P offer-book listings.
Reental	Real Estate	P2P (centralized marketplace listings; direct OTC transfers)	Real estate tokenization platform focusing on rental real estate and pre-developments in Spain and Latin America with platform-gated secondary trading.
Figure	Private Credit / HE-LOCs	AMM/Pool-based (yield/loan pools) / Alternative Trading Systems (ATS)	Publicly listed tokenization platform specialized in tokenized consumer loans, most notably Home Equity Lines of Credit (HELOCs). Provides ATS trading via Figure Markets.
Maple Finance	Private Credit	AMMs/Pool-based (underwritten lending pools, DeFi wrapper)	On-chain institutional private credit via curated lending pools. Liquidity is primarily via pool withdrawals; secondary exit liquidity is limited.
Centrifuge	Private Credit	AMMs/Pool-based (liquidity pools)	Tokenizes real-world credit, invoices and receivables for DeFi lending; decentralized credit pools.
Artory	Art	Secondary trading via Tokeny Integration	Partners with tokenization infrastructure providers to enable compliant secondary trading for traditionally illiquid art market on Ethereum.
Aktionariat	Private Equity / Stocks	On-chain order book / AMM/Pool-based (Hybrid)	Swiss platform allowing direct trading of tokenized company shares (AGs) using blockchain infrastructure.
Swarm Markets	Stocks / US Treasuries / Gold	AMM/Pool-based (Balancer-fork) / P2P ("dOTC", decentralized OTC)	German (BaFin)-regulated DeFi platform enabling compliant trading via AMM-based DEX and on-chain OTC service.
Securitize	Private Funds / Stocks	Centralized matching / Secondary market (ATS) / AMM/Pool-based (UniswapX)	SEC-registered platform offering third-party tokenized securities and funds with secondary trading via regulated ATS and DeFi integrations.
Ondo Finance	Stocks / ETFs / Bonds	Buyback/Redemption (24/5 redemption linked to public markets) / AMM/Pool-based	Institutional-grade RWA issuer offering tokenized Treasuries and listed stocks/ETFs with redemption options linked to underlying public markets and DeFi composability.

Note: This table provides a non-exhaustive overview of selected RWA tokenization platforms and the secondary-market liquidity mechanisms they employ. Platforms are classified into three broad mechanism categories: **AMM/Pool-based** mechanisms, which include AMMs with algorithmic pricing and continuous quoting and other pool-based capital allocation vehicles; **P2P** mechanisms based on bilateral or posted-offer trading; and **Buyback/Redemption** mechanisms offering issuer-provided liquidity at appraisal- or market-linked prices. Several platforms combine multiple mechanisms. Platform descriptions are based on publicly available documentation and observed market functionality.

A.1.1. RWA marketplaces and institutional details

AMM mechanism. AMMs (Figure A.1) provide continuous, two-sided liquidity through algorithmic pricing based on liquidity pools. Trades execute immediately at prices determined by the CPMM formula, implying convex price impact that increases with trade size. In RealT’s AMMs, LPs earn trading fees and continue to receive rental income on the RWA tokens they contribute. In addition, the issuer seeds each pool with retained tokens and redistributes the associated cash flows to external LPs pro rata. This creates a declining marginal subsidy: external LP yields are highest when the pool is small and fall as participation deepens.

Figure A.1: AMM mechanism



Note: This figure illustrates the schematic workflow of an automated market maker (AMM) mechanism.

Levenswap, the dominant AMM venue in our setting, implements the Uniswap V2 constant-product AMM design (Adams et al., 2020). AMMs offer high execution certainty and low fees but expose traders and LPs to slippage and inventory risk when liquidity is thin and paired pool tokens are uncorrelated. Arbitrageurs continuously adjust pool prices in response to order flow, linking AMM prices to other market signals (Angeris et al., 2019). Liquidity pools are typically created shortly after a RWA STO. Liquidity provision and withdrawal events in constant-product AMMs do not mechanically alter price signals, as reserves are adjusted proportionally on both sides of the pool.

AMM price signals and liquidity pools. We extract liquidity-pool reserve states before and after each executed token amount (swap) from Uniswap (V1/V2 on Ethereum) and Levenswap (Gnosis). Each RWA token is associated with a distinct liquidity pool, typically paired with a stablecoin quote token, most frequently USDC. Table A.2 reports the distribution of all quote-token regimes across the sample. The regimes are chronologically ordered from top to bottom: ETH and WETH on Ethereum until March 2021, LEVIN from March 2021 until

its collapse later that year, WETH on Gnosis as an interim quote token, and USDC as the dominant quote token from May 2022 onward.

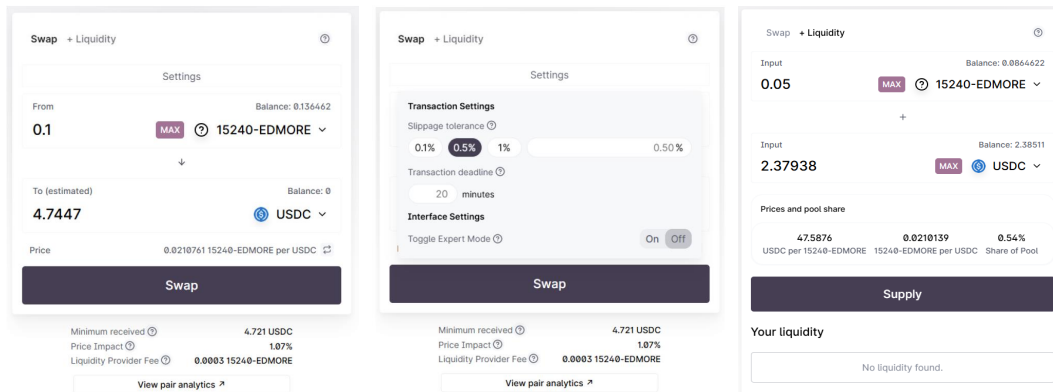
Table A.2: AMM event counts by pool quote token

Pool quote token	N	Swap	Deposit	Withdrawal	Total
ETH	9	5,622	286	182	6,090
LEVIN	98	161,148	3,016	1,444	165,608
WETH	98	294,704	1,253	547	296,504
USDC	560	93,700	3,985	2,016	99,701
Total	765	555,174	8,540	4,189	567,903

Note: This table reports AMM event counts (*Swap*, *Deposit*, and *Withdrawal*) by pool quote token. *ETH* (Ether) is Ethereum’s native token, *WETH* (Wrapped Ether) is the ERC-20 wrapped representation of ETH used across Ethereum-compatible blockchains and applications, *LEVIN* is the Levinswap protocol token, and *USDC* is the USD-pegged stablecoin (USD Coin). *N* is the number of distinct liquidity pools using the respective quote token. The slight differing number of pools (765) and issued RWA tokens (752) is driven by partially duplicated pools from the Ethereum-to-Gnosis migration and by RWA tokens with pools but limited trade activity at the time of data collection.

Pool-implied prices are computed using the CCPM rule, where the price equals the ratio of quote-token reserves to base-token reserves, i.e., $P = y/x$, with x denoting the reserve quantity of the RWA token (base-token) and y the reserve quantity of the quote token. This construction yields a continuous AMM price signal that can be matched to AMM, P2P, and Buyback transactions. For AMM trades, we use the pre-swap reserve state implied by the executed token amounts to avoid mechanical endogeneity. Slippage is defined as the relative deviation between the execution price and the pre-swap pool price, measured as $(P_{\text{Execution}} - P_{\text{Before}})/P_{\text{Before}}$, where $P_{\text{Execution}}$ is the realized swap price and P_{Before} is the pool-implied price based on reserves immediately prior to execution.

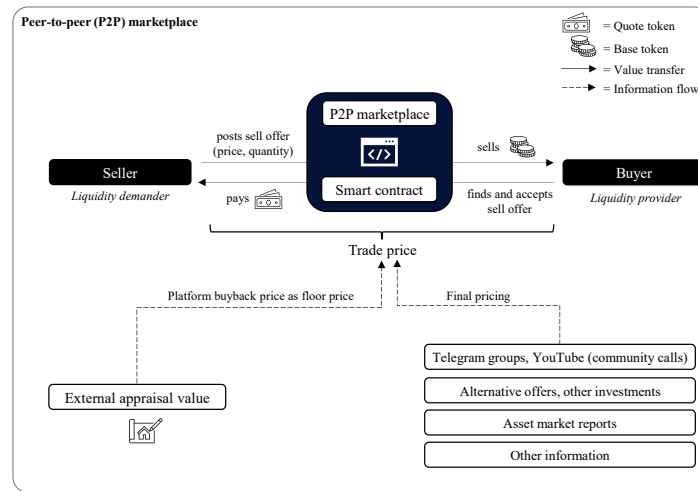
Figure A.2: AMM: Levinswap web interface



Note: The left and the middle screenshots show the Levinswap web interface for an AMM swap execution (trade), and the right screenshot shows the interface for liquidity provision. Accessed via <https://app.levinswap.realtoken.community/> on Oct 29, 2025.

P2P mechanism. P2P trading (Figure A.3) occurs via posted-offer marketplaces that allow users to specify prices and quantities directly. In our setting, trading takes place on the Swapcat and YAM marketplaces and almost exclusively on Gnosis. The marketplaces support buy, sell, and exchange orders through smart contracts. Offers become visible shortly after confirmation and remain active until matched, updated, or canceled. Execution is stochastic and depends on search frictions and posted depth, with limited buy-side liquidity and bid–ask spreads that remain persistent when two-sided quotes are posted.

Figure A.3: Peer-to-peer (P2P) marketplace



Note: This figure illustrates the transaction process on a smart-contract-based P2P market design.

YAM was introduced in December 2022 and provides a searchable live offer board with detailed metadata, including trade direction, price, quantity, implied yield, and the contemporaneous Buyback valuation. Private offers can be arranged here. Swapcat offers a simpler interface without private offers. Both platforms record and settle offers and transactions on-chain. Overall, the observed P2P marketplaces resemble decentralized OTC markets with flexible pricing and uncertain execution.

Figure A.4: P2P: YAM web interface

Offer ID	Offer Token	Buyer Token	Offer Yield	Other Yield	Yield Delta	Offer Price	Price Delta	Available Quantity
117585	Redwood PR 6 501 - OF	RealT Token (USD)	11.00%	11.00%	-0.00%	\$0.80 USD	\$0.84 (\$0.80)	0.000000000000000000
117511	16167 Greenfield - OF	USD-IC on-chain	11.00%	10.00%	+10.00%	\$0.84 USD	\$0.84 (\$0.79)	0.0
117517	2 Holdings Palace - OF	USD-IC on-chain	10.00%	10.00%	-0.00%	\$0.21 USD	\$0.19 (\$0.18)	0.000000000000000000
117514	1 Holdings Palace - OF	USD-IC on-chain	10.00%	9.20%	+8.00%	\$0.89 USD	\$0.45 (\$0.44)	0.000000000000000000
117527	16028 Palace - OF	USD-IC on-chain	11.20%	10.70%	+0.50%	\$0.81 USD	\$1.00 (\$0.80)	2
117564	5231 Longene - OF	USD-IC on-chain	10.00%	10.00%	-0.00%	\$0.00 USD	\$0.00 (\$0.00)	2
117565	15400 Palace - OF	USD-IC on-chain	10.00%	10.00%	-0.00%	\$1.00 USD	\$1.00 (\$0.99)	1.000000000000000000
117558	14128 Stale - OF	USD-IC on-chain	8.11%	8.00%	+0.10%	\$0.45 USD	\$0.84 (\$0.00)	0.000000000000000000
117559	14128 Stale - OF	Whispered KDA	8.11%	8.00%	+0.10%	\$0.45 USD	\$0.84 (\$0.00)	0.000000000000000000
117559	14128 Stale - OF	RealToken Swapcat (USD)	8.11%	8.00%	+0.10%	\$0.45 USD	\$0.84 (\$0.00)	0.000000000000000000

Note: This figure illustrates the web interface to explore current offers on YAM. Accessed via <https://yam.realtoken.network/> on Oct 29, 2025.

P2P price signals and offer-book depth. P2P price signals are inferred from the reconstruction of the complete on-chain offer histories of the two active P2P marketplaces, Swapcat and YAM, and merged together on the token-level and under the application of specific realism filters, explained below. For each marketplace, we recover the full life cycle of every offer, including creation, updates, partial and full executions, and cancellations, using marketplace-specific parsing routines. On YAM, both prices and quantities can be updated, whereas on Swapcat only price updates are permitted. For offer creations, updates, and cancellations usual blockchain gas fees apply. Price signals are split into bid and ask quotes based on offer direction, and reflect the contemporaneous, publicly visible offer inventory. For multiple offers for the same token at the same time, we apply the median and a liquidity-weighted aggregation, weighted by offer depth. Table A.3 reports daily summary statistics for realistic P2P depth and price measures at both the single-token and aggregate-market level. For each executed P2P trade, we compute *time on market* as the elapsed time between offer creation and execution. While this metric could alternatively be measured from the most recent price or quantity update, we consistently use *time on market since offer creation (days)* as the primary time-to-execution measure throughout the execution-cost framework and empirical analysis.

Realism filters. To ensure that P2P depth and price signals reflect executable and economically meaningful liquidity, we apply a sequence of realism filters. We exclude *private offers*, which are only available on YAM (3,951 offers), as they are not publicly observable.

We further remove offer states with zero *remaining inventory* (3,991 offers), as well as stale liquidity.

An offer-book state is classified as *stale* if its effective time on market, defined as the minimum of time since creation and time since last update, exceeds 60 days. Across marketplaces and sides, the 90th percentile of time since creation is approximately 49 days and the 90th percentile of time since last update is approximately 17 days, placing states persisting beyond 60 days in the extreme upper tail and unlikely to execute. In offer-state space, these screens remove 10,836 private states, 241,616 zero-inventory states, and 59,160 stale states.

Finally, we filter unrealistically priced offers by benchmarking P2P quotes against the concurrent Buyback price. The 90th percentile of the absolute price deviation is 24.6% on the ask side and 25.8% on the bid side; offers with deviations exceeding 30% are classified as *out-of-the-money* and removed (16,259 offers corresponding to 45,942 offer states).

On July 22, 2025, RealT temporarily suspended its Buyback for affected property tokens due to pending litigation; the same realism filters continue to apply during this period.

Overall, these filters remove approximately 14% of originally observed P2P offer-book depth when measured in USD inventory. Across the full sample of 777,038 transactions, this procedure yields 635,475 realistic ask-side and 167,381 realistic bid-side P2P price signals. The first P2P transaction occurs on October 26, 2020 on Swapcat, but activity increases substantially following the launch of YAM in early 2023. As a result, the majority of concurrent P2P price signals for non-P2P transactions are observed from 2023 onward (510,110 ask-side and 105,099 bid-side observations), with bid-side depth remaining sparse prior to the end of 2024, as illustrated in Figure 7.

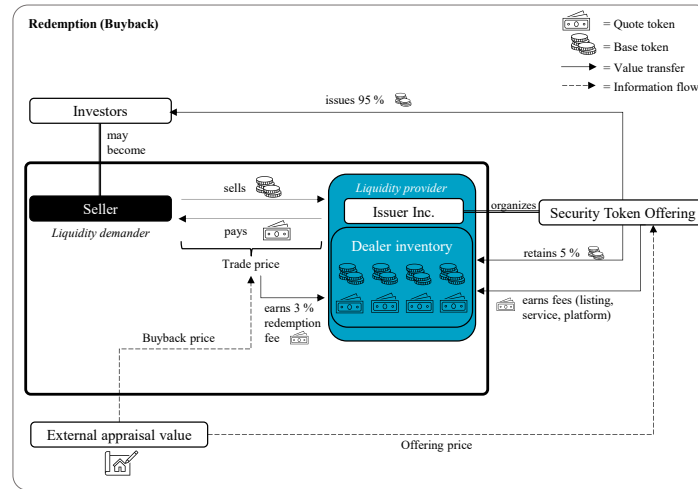
Table A.3: Realistic P2P price signals order-book depth measures

Statistic	N	Mean	St. Dev.	Min	10%	Median	90%	Max
<i>One RWA token</i>								
Bid offer depth (USD)	1,671	186.74	455.96	0.00	0.00	0.00	574.89	3,034
Ask offer depth (USD)	1,671	932.89	1,156	0.00	58.00	597.04	2,029	15,069
Number of bid offers	1,671	1.25	2.50	0.00	0.00	0.00	5.00	14.00
Number of ask offers	1,671	3.04	2.18	0.00	1.00	2.50	6.00	12.00
Price signal bid offer (median, USD)	1,041	51.31	4.14	40.00	49.02	51.81	53.80	142.86
Price signal ask offer (median, USD)	1,629	56.04	7.87	45.99	52.46	55.42	58.00	168.00
Price signal bid offer (liquidity-weighted, USD)	1,041	51.38	4.10	40.00	49.16	51.87	53.88	142.86
Price signal ask offer (liquidity-weighted, USD)	1,629	56.43	7.84	45.99	53.02	55.88	58.46	168.00
<i>All RWA tokens</i>								
Bid offer depth (USD)	1,671	105,135	231,234	0.00	0.00	8,089	415,888	1,550,938
Ask offer depth (USD)	1,671	473,837	540,130	0.00	1,704	285,429	1,130,408	3,319,229
Number of bid offers	1,671	465.49	1,037	0.00	0.00	43.50	1,932	7,016
Number of ask offers	1,671	728.19	1,139	0.00	2.00	410.50	1,500	8,034
<i>Total activity counts</i>								
Total offers created								179,771
Total offer updates								127,867
Total purchases								207,885

Note: This table summarizes reconstructed realistic P2P price signals and offer-book depth after applying the realism filters defined in the accompanying Appendix text. For each token-day, offer-book measures are computed as the median across all observed offer events (offer creations, updates, purchases, and removals) within that token and day. *One RWA token* rows report summary statistics across days of the daily cross-sectional median across tokens. *All RWA tokens* rows report summary statistics across days of daily sums across tokens (depth and offer counts). Depth is measured in USD. Price statistics are computed only on days with at least one realistic quote on the corresponding side, so *N* may be below the number of calendar days.

Buyback mechanism. The Buyback mechanism (Figure A.5) provides a centralized issuer-provided liquidity outlet in which investors sell tokens back to the issuer at an appraisal-based price. Pricing is stable but infrequently updated, and execution is subject to quantity caps and deterministic delay (up to 10 working days). At RealT, Buyback is capped at \$2,000 per wallet per week, requires that at least 95% of tokens are sold to investors, and settles with a deterministic waiting period of up to ten business days. Transactions settle on-chain in USDC via Ethereum or Gnosis and incur blockchain gas fees and a 3% platform redemption fee.

Figure A.5: Buyback



Note: This figure illustrates the Buyback process as a centralized market design.

Buyback price signals. Buyback price signals are based on third-party property appraisal values and retrieved from the [RealT API](#). Using the full history of Buyback prices and their associated revaluation dates, we assign a contemporaneous Buyback price signal to every non-Buyback transaction. Because the Buyback mechanism was the first form of secondary trading available for these tokens, Buyback price signals are available without gaps across the entire non-Buyback transaction sample, yielding 777,038 observations.

STO process Our dataset consists of real estate tokens issued by RealToken LLC (RealT), a platform specializing in the tokenization of rental residential real estate. Under the Howey framework, these tokens qualify as investment contracts and are therefore regulated as securities under US law. RealT initially conducted STOs under Regulation D 506(c) for US-accredited investors, but as of October 29, 2025, all active real estate tokens have been converted to Regulation S, and US investors are no longer permitted to invest in RealT real estate tokens.¹ For further details on the STO process, see [Kreppmeier et al. \(2023\)](#).

¹See <https://faq.realt.co/en/article/what-is-realt-who-can-invest-how-do-i-invest-11yc5h5/>.

Each tokenized property is held by a dedicated special purpose vehicle (SPV), RealToken Series LLC, which holds legal title to the property and operates as a standalone legal entity. Ownership interests in these SPVs are represented by ERC-20 tokens issued on Ethereum and Gnosis. The underlying assets are located predominantly in the US with an increasing number abroad, mainly Latin America. Property management is predominantly outsourced to local operators. Investors acquire tokens during the STO after completing payment and digitally signing the offering memorandum, after which tokens are transferred to investor wallets via smart contracts. For most tokens, ownership represents an equity claim on the corresponding SPV, and net rental income, after operating expenses, insurance, and taxes, is distributed weekly to token holders through smart contracts. The RealT ecosystem further includes a decentralized autonomous organization (DAO) through which token holders can vote on specified governance matters, such as approving de-tokenizations following purchase offers for the underlying properties.

A.2. Stylized execution cost framework

Table A.4: Notation summary for the execution cost framework

Symbol	Definition	Specification / Notes
Prices and quantities		
P	Reference price (USD) for normalizing mechanism costs	Costs expressed relative to P
$q > 0$	Trade size (tokens), buy or sell, fractional allowed	Units of the base token
Fees		
F_A	AMM trading fee (relative)	Applied to notional qP
F_P	P2P trading fee (relative)	Applied to notional qP
$F_B = \frac{\Delta P_B}{P}$	Buyback discount relative to reference price	$\Delta P_B = P - P_B$; captures appraisal-based haircut plus redemption fee charged by issuer
Liquidity variables		
L_A	AMM liquidity (base-token units in pool)	Determines AMM slippage via constant-product rule
L_P	P2P liquidity (token units on relevant book side)	Ask depth for buys, bid depth for sells
Time and preference parameters		
τ_B	Expected time to execution in Buyback	Deterministic waiting time
$c\tau_B$	Expected P2P execution time (time on market) when $L_P = 0$	$c \neq 1$ allows faster/slower matching than Buyback
$\lambda \in [0, 1]$	Time preference / urgency parameter	Higher λ penalizes delay more strongly
Price-impact terms		
PI^0	Expected P2P price impact when $L_P = 0$	Bounded above by F_B ; thin-market concession
PI^+	Expected P2P impact when $q = L_P > 0$	Benchmark for depth-based impact calibration ("walking the book")
$\delta \in (0, 1)$	Curvature parameter for sub-quadratic P2P impact	Implies impact exponent $1 + \delta \in (1, 2)$

Note: This table summarizes the notation used in the mechanism choice. All costs are expressed in units of qP and therefore normalized with respect to the reference price P . P2P liquidity L_P refers to the standing depth on the relevant side of the offer book (ask-side offers to sell and bid-side offers to buy). The impact parameters PI^0 , PI^+ , and δ govern price concessions in thin or shallow P2P markets, while AMM slippage is determined by the constant-product pricing rule through L_A . Time-to-execution enters linearly in expected delay costs through λ . All functional forms below should be interpreted as reduced-form approximations calibrated directly from the data (fees, depth, execution time) rather than structural primitives.

Trader preferences and choice.

Following conventions and DeFi AMM studies (e.g., [Capponi and Jia, 2021](#)), we treat the RWA token as the base asset or token and the stablecoin as the quote asset or token. Prices are quoted in units of the quote asset.

Mechanism-specific execution costs. All three mechanisms share the same reference price P but differ in fee schedules, price impact, and time-to-execution. Each total cost expression contains the factor $qP > 0$; when comparing mechanisms, we divide by qP and work with normalized costs.

AMM. For an AMM with base-token depth L_A and reference price P , the post-trade price after a trade of size q is

$$P_{\text{new}} = \begin{cases} P \left(\frac{L_A}{L_A - q} \right)^2 & (\text{buy}), \\ P \left(\frac{L_A}{L_A + q} \right)^2 & (\text{sell}). \end{cases} \quad (18)$$

The relative slippage is therefore $\left(\frac{L_A}{L_A - q} \right)^2 - 1$ for buys and $1 - \left(\frac{L_A}{L_A + q} \right)^2$ for sells. Since the trader executes along the AMM price curve, we proxy the average execution-price impact by one half of the marginal price change from P to P_{new} , which is the convention in AMM trade settlements but also holds approximately when settling a large trader though of various small orders in P2P markets. Normalized AMM costs are

$$\begin{aligned} c_A^{\text{buy}}(q; L_A) &:= \frac{C_A^{\text{buy}}}{qP} = F_A + \frac{1}{2} \left[\left(\frac{L_A}{L_A - q} \right)^2 - 1 \right], \\ c_A^{\text{sell}}(q; L_A) &:= \frac{C_A^{\text{sell}}}{qP} = F_A + \frac{1}{2} \left[1 - \left(\frac{L_A}{L_A + q} \right)^2 \right]. \end{aligned} \quad (19)$$

The fee term $F_A qP$ is an approximation, because the actual fee depends on the realized transaction price. The error is second-order relative to slippage and empirically negligible in our setting.

Buyback. The Buyback executes at a fixed discount F_B but with deterministic delay τ_B . We assume delay costs scale with invested volume qP , so the normalized Buyback cost is

$$c_B := \frac{C_B}{qP} = F_B + \lambda \tau_B. \quad (20)$$

P2P with no standing depth ($L_P = 0$). When the relevant side of the P2P book is empty, the trader faces search frictions and a thin-market concession. We summarize the expected price concession as PI^0 and the expected execution time as $c\tau_B$ (with $c \neq 1$ allowing for faster or slower matching than the Buyback). Normalized P2P costs are

$$c_P^{(0)} := \frac{C_P^{(0)}}{qP} = F_P + PI^0 + \lambda c \tau_B. \quad (21)$$

A natural upper bound on PI^0 in case of a sell order is the Buyback discount F_B .

P2P with depth ($L_P > 0$). When there is standing depth $L_P > 0$ on the relevant side of the P2P book, we assume sub-quadratic price impact, consistent with the empirical literature

(Almgren and Chriss, 2001) and market-microstructure intuition of “walking the book”:

$$c_P^{(+)}(q; L_P) := \frac{C_P^{(+)}}{qP} = F_P + \frac{1}{2} PI^+ \left(\frac{q}{L_P} \right)^\delta, \delta \in (0, 1). \quad (22)$$

This implies impact $\propto q^{1+\delta}$, with exponent between linear and quadratic.

Routing for buy orders.

For buy orders, the Buyback does not exist in our empirical setting, so we focus on AMM vs. P2P. The trader compares c_A^{buy} and $c_P^{(+)}$ when $L_P > 0$ and c_A^{buy} and $c_P^{(0)}$ when $L_P = 0$.

Main case: $L_P > 0$.

Proposition 1. A buy order of size q is routed to P2P (with $L_P > 0$) if

$$\left(\frac{L_A}{L_A - q} \right)^2 - 1 - PI^+ \left(\frac{q}{L_P} \right)^\delta > 2F_P - 2F_A, \quad (23)$$

and to the AMM otherwise.

Minor case: $L_P = 0$.

Proposition 2. A buy order of size q is routed to P2P (with $L_P = 0$) if

$$\lambda < \frac{1}{c\tau_B} \left(\frac{1}{2} \left[\left(\frac{L_A}{L_A - q} \right)^2 - 1 \right] + F_A - PI^0 - F_P \right), \quad (24)$$

and to the AMM otherwise.

Routing for sell orders.

For sell orders, all three mechanisms are potentially relevant. The trader compares c_A^{sell} , c_B , and $c_P^{(0)}$ or $c_P^{(+)}$ depending on L_P .

Condition 1. Given a sell request (q, λ) , the AMM is preferred to the Buyback if

$$\lambda < \frac{1}{\tau_B} \left(F_A - F_B + \frac{1}{2} - \frac{1}{2} \left(\frac{L_A}{L_A + q} \right)^2 \right). \quad (25)$$

Condition 2. Given a sell request (q, λ) and $L_P = 0$, P2P is preferred to Buyback if

$$\lambda < \frac{1}{\tau_B(c-1)} \left(F_B - PI^0 - F_P \right) \quad \text{if } c > 1, \quad (26)$$

and

$$\lambda > \frac{1}{\tau_B(c-1)} \left(F_B - PI^0 - F_P \right) \quad \text{if } c < 1. \quad (27)$$

Condition 3. Given a sell request (q, λ) and $L_P > 0$, P2P is preferred to Buyback if

$$\lambda > \frac{1}{\tau_B} \left(F_P - \frac{1}{2} P I^+ \left(\frac{q}{L_P} \right)^\delta - F_B \right). \quad (28)$$

Condition 4. Given a sell request (q, λ) and $L_P = 0$, P2P is preferred to the AMM if

$$\lambda < \frac{1}{c\tau_B} \left(F_A + \frac{1}{2} - \frac{1}{2} \left(\frac{L_A}{L_A + q} \right)^2 \right). \quad (29)$$

Condition 5. Given a sell request (q, λ) and $L_P > 0$, P2P is preferred to the AMM if

$$1 - \left(\frac{L_A}{L_A + q} \right)^2 - P I^+ \left(\frac{q}{L_P} \right)^\delta > 2F_P - 2F_A. \quad (30)$$

The inequalities in Conditions 1–5 partition (q, λ) -space into regions where AMM, P2P, or Buyback is optimal, conditional on $(L_A, L_P, F_A, F_P, F_B, \tau_B, c, P I^0, P I^+, \delta)$.

Comparative statics in liquidity

Proposition 3. All else equal, AMM becomes more preferable over P2P when L_A increases, while P2P becomes more attractive when L_P increases.

Proof. Consider the buy case with $L_P > 0$ and define

$$\Delta(L_A, L_P; q) := \left(\frac{L_A}{L_A - q} \right)^2 - 1 - P I^+ \left(\frac{q}{L_P} \right)^\delta. \quad (31)$$

Then

$$\frac{\partial}{\partial L_A} \left(\frac{L_A}{L_A - q} \right)^2 = \frac{-2L_A q}{(L_A - q)^3} < 0 \quad (32)$$

for $L_A > q$, and

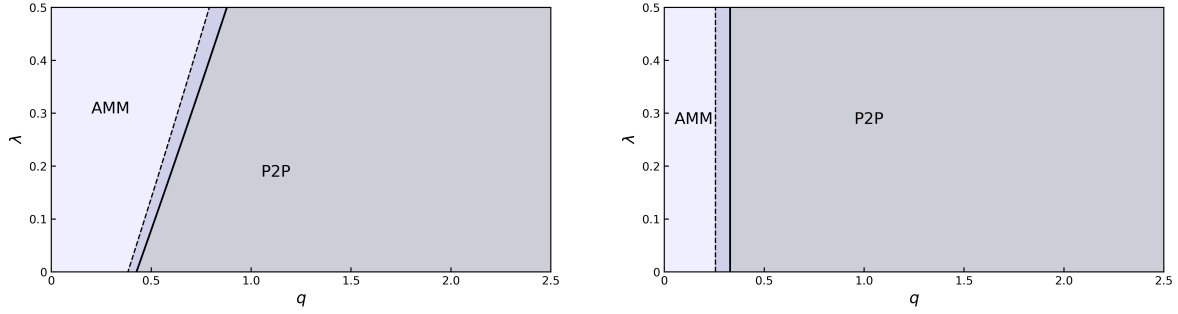
$$\frac{\partial}{\partial L_P} \left[-P I^+ \left(\frac{q}{L_P} \right)^\delta \right] = P I^+ \delta q^\delta L_P^{-\delta-1} > 0. \quad (33)$$

Thus $\partial\Delta/\partial L_A < 0$ and $\partial\Delta/\partial L_P > 0$. Since P2P is chosen whenever $\Delta(L_A, L_P; q) > 2F_P - 2F_A$, a c.p. increase in L_A or a c.p. decrease in L_P lowers the advantage of P2P and shifts routing toward the AMM. On the sell side, the corresponding term replaces $\left(\frac{L_A}{L_A - q} \right)^2 - 1$ with $1 - \left(\frac{L_A}{L_A + q} \right)^2$; the derivative signs are unchanged.

Illustrative example

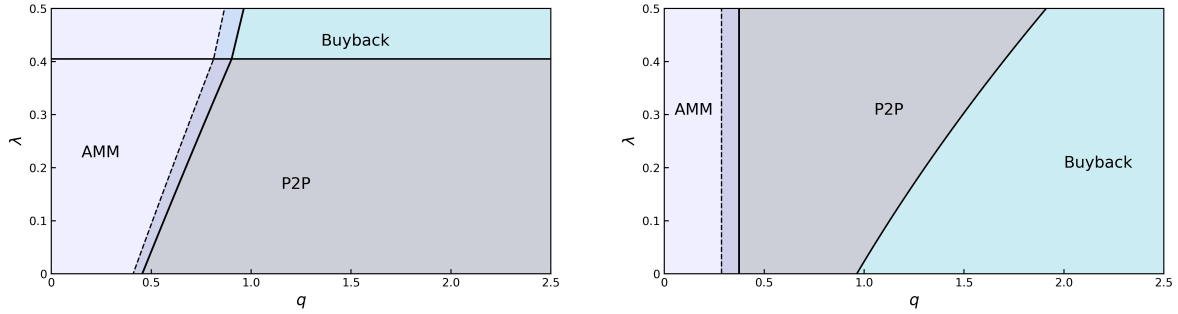
Figures A.6 and A.7 illustrate the implied routing regions for realistic parameter values, which can be observed in our data: $F_B = 0.03$, $F_A = 0.003$, $F_P = 0$, $L_P = 3$ (if not 0), $L_A = 20$, $\tau_B = \frac{1}{36}$ (10 days), $c = 1.8$, $P I^0 = 0.025$, $P I^+ = 0.12$, and $\delta = 0.5$.

Figure A.6: Buy order: rational choice between AMM and P2P (left: $L_P = 0$, right: $L_P > 0$)



Note: Dashed lines indicate a decrease in L_A by two tokens.

Figure A.7: Sell order: rational mechanism choice (left: $L_P = 0$, right: $L_P > 0$)



Note: Dashed lines indicate a decrease in L_A by two tokens.

A.2.1. AMM liquidity provision framework

In each liquidity pool, the RWA token is the base asset and a cryptocurrency token (typically a USD stablecoin) is the quote asset. The token initiator seeds the pool with L_R base tokens and the corresponding amount of the quote token at price P (initially equal to the Buyback price). An external LP chooses how many additional tokens $L > 0$ to contribute, along with the matched quote amount, so that the total base-token depth in the pool is $L_R + L$.

The LP keeps this position in the AMM for a fixed horizon (e.g. one year). Over this horizon, the pool faces a total sell volume q_s (base tokens sold into the pool) and buy volume q_b (base tokens bought out of the pool). We define

$$\Delta q := q_b - q_s, \quad Q := q_b + q_s. \quad (34)$$

Here, Δq is the net order imbalance (positive when buy pressure dominates), and Q is the total trading activity. At the end of the period, the base-token inventory in the pool equals $L_R + L - \Delta q$. Ex ante, Q and Δq are unknown and must be forecast by the LP.

The LP's profit over the horizon has four components:

1. Rental income on the tokens in the pool, i.e. the yield from the token held in the pool ($L\mu P$) plus the sponsored yield from the token initiator ($L_R\mu P$), resulting in the total yield component of:

$$\left(L + L_R - \frac{\Delta q L}{2(L_R + L)}\right) \mu P \quad (35)$$

(the subtrahend corrects for the varying number of tokens held during the interval, μ denotes the rental yield).

2. Trading fee income proportional to the total traded number Q amounting to

$$Q F_A \frac{L}{L_R + L} P. \quad (36)$$

3. The valuation changes arising from the constant-product pricing rule. Initially an amount of capital of $2P(L_R + L)$ is invested in the liquidity pool, which after the period amounts to

$$2P(L_R + L) \frac{L_R + L}{L_R + L - \Delta q}. \quad (37)$$

Therefore, for the LP the profit/loss from this valuation change is:

$$2P(L_R + L) \frac{\Delta q}{L_R + L - \Delta q} \frac{L}{L_R + L} = 2PL \frac{\Delta q}{L_R + L - \Delta q}. \quad (38)$$

This corresponds to the LP's fraction of the total profit in the liquidity pool. However, for reasons of mathematical tractability, we first approximate this term by linearizing it

$$2PL \frac{\Delta q}{L_R + L}. \quad (39)$$

Note that for small values of Δq (against $L_R + L$), which are realistic in our context, the approximation makes only a negligible error.

4. The opportunity cost of capital used to finance the AMM position.

$$-2rPL. \quad (40)$$

Collecting terms and normalizing by P , the expected profit can be written as

$$\Pi(L) = \frac{aL^2 + bL + c}{L_R + L}, \quad (41)$$

with

$$\begin{aligned} a &= \mu - 2r, \\ b &= 2L_R(\mu - r) + QF_A + \Delta q\left(2 - \frac{\mu}{2}\right), \\ c &= \mu L_R^2. \end{aligned} \tag{42}$$

where μ is the yield earned by holding the base token, r is the LP's cost of capital, and F_A is the AMM fee rate. We focus on the empirically relevant case $2r > \mu$, under which $\Pi(L)$ is concave in L .

External AMM liquidity

Let the “revenue intensity” be $\gamma := QF_A + \Delta q\left(2 - \frac{\mu}{2}\right)$. If $2r > \mu$ and $\gamma > 0$, the LP's profit-maximizing external liquidity is

$$L^* = -L_R + \sqrt{L_R \frac{\gamma}{2r - \mu}}. \tag{43}$$

Obviously, optimal AMM depth increases with γ , which captures expected fee income QF_A and expected gains from net buy imbalance Δq . When expected revenues are low relative to the capital cost spread $2r - \mu$, optimal external liquidity is small or in many realistic cases even negative, i.e. the optimum value cannot be achieved and the marginal optimum is $L = 0$. Thus, for empirical implementation, it is more relevant to characterize the range of L for which liquidity provision is profitable at all.

Profitability region and the role of L_R

Suppose $2r > \mu$ and define γ as above. Then analyzing the numerator of (41) with respect to zeros is a straightforward quadratic problem resulting in the LP's profit being non-negative for all $L \in [0, L']$ with

$$\begin{aligned} L' &= \frac{1}{2(2r - \mu)} \left[2L_R(\mu - r) + \gamma \right. \\ &\quad \left. + \sqrt{4L_R^2 r^2 + 4L_R(\mu - r)\gamma + \gamma^2} \right]. \end{aligned} \tag{44}$$

Moreover, $\partial L' / \partial L_R > 0$ implies higher initiator seed liquidity L_R strictly enlarges the range of profitable external liquidity provision. Note that for $r = \mu$ and an estimated $\gamma = 0$ (which can approximate a thin trading activity and balanced buy and sell orders), we have $L' = L_R$.

Equation (44) delivers a direct link between initiator commitment and sustainable AMM depth: the more tokens the initiator seeds, the larger the set of $(Q, \Delta q)$ realizations for which external LPs can earn non-negative profits. In our empirical tests, this prediction translates into higher observed AMM depth and higher AMM usage when L_R is high.

Differentiating (44) with respect to L_R yields

$$\begin{aligned} \frac{\partial L'}{\partial L_R} = \frac{1}{2(2r - \mu)} & \left[2(\mu - r) \right. \\ & \left. + \frac{4L_R r^2 + 2(\mu - r)\gamma}{\sqrt{4L_R^2 r^2 + 4L_R(\mu - r)\gamma + \gamma^2}} \right]. \end{aligned} \quad (45)$$

If we define $D := 4L_R^2 r^2 + 4L_R(\mu - r)\gamma + \gamma^2$, then

$$\frac{\partial L'}{\partial L_R} = \frac{2(\mu - r)\sqrt{D} + 4L_R r^2 + 2(\mu - r)\gamma}{2(2r - \mu)\sqrt{D}}. \quad (46)$$

Here, the denominator is positive because of $2r > \mu$. To see the positivity of the numerator of (46), consider that D can be written as

$$D = (\gamma + 2L_R(\mu - r))^2 + 4L_R^2 \mu(2r - \mu).$$

Thus $\sqrt{D} \geq |\gamma + 2L_R(\mu - r)|$, which implies (as a weaker inequality) $\sqrt{D} + \gamma \geq -2L_R(\mu - r)$. The numerator of (46) can be rewritten as

$$N := 2(\mu - r)(\sqrt{D} + \gamma) + 4L_R r^2.$$

Using the inequality regarding $\sqrt{D} + \gamma$, we have

$$N \geq 2(\mu - r)(-2L_R(\mu - r)) + 4L_R r^2 = 4L_R \mu(2r - \mu) > 0 \quad (47)$$

Altogether the claim $\partial L' / \partial L_R > 0$ is proven.

A.3. Definition of all variables

Table A.5: Definition of all variables

Variable	Definition	Calculation
Secondary-market variables (intraday perspective)		
Trades and price signals		
<i>Trade size (RWA token)</i>	Absolute number of RWA tokens transferred in the trade (base-token amount; for AMM trades: the exact token amount swapped in the pool).	Own
<i>Trade volume (USD)</i>	USD notional of the trade (quote-leg value): the quote-token amount exchanged, converted to USD at the trade timestamp (for AMM trades: the exact quote-token amount swapped in the pool, valued in USD).	Own
<i>Total fees (USD)</i>	Total fees paid in a trade, converted to USD: blockchain transaction fees (all mechanisms), issuer-platform fee on Buyback trades (3% of volume), and AMM trading fees on AMM trades (0.3% of volume, split into 0.25% to LPs and 0.05% to the protocol).	Own

<i>Trade price (USD/RWA token)</i>	Execution price per RWA token, expressed in USD. For AMM trades, computed from on-chain swap amounts and reserve states.	Own
<i>RWA token buy</i>	Dummy equal to one for buyer-initiated trades (purchase of RWA token), and zero for seller-initiated trades.	Own
<i>Ethereum blockchain</i>	Dummy equal to one if the trade executes on Ethereum, and zero if on the Gnosis.	Own
<i>Price signal Buyback (USD)</i>	Buyback price (USD per RWA token) quoted by the issuer platform at the time of the trade.	RealT
<i>Price signal AMM (USD)</i>	AMM-implied price (USD per RWA token) computed from pool reserves at the time of a P2P/Buyback trade, or immediately prior to an AMM trade.	Own
<i>Price signal P2P bid offer (USD)</i>	Posted bid price (USD per RWA token), i.e., willingness to buy the RWA token for a posted quote-token amount. If multiple bids are available at trade time, the median bid price is used.	Own
<i>Price signal P2P ask offer (USD)</i>	Posted ask price (USD per RWA token), i.e., willingness to sell the RWA token for a posted quote-token amount. If multiple asks are available at trade time, the median ask price is used.	Own

AMM and P2P liquidity

<i>AMM pool liquidity bid (USD)</i>	USD value of the quote-token reserve in the AMM pool at trade time (capacity to buy RWA tokens).	Own
<i>AMM pool liquidity ask (USD)</i>	USD value of the RWA-token reserve in the AMM pool at trade time (capacity to sell RWA tokens), valued at the pool-implied price.	Own
<i>Total AMM pool liquidity (USD)</i>	Total AMM pool value (USD) at trade time, defined as the sum of the USD values of both reserves.	Own
<i>Deposit (USD)</i>	Quote-token amount deposited into the AMM liquidity pool (USD), paired with an RWA-token deposit in the prevailing reserve ratio.	Own
<i>Deposit (RWA token)</i>	RWA-token amount deposited into the AMM liquidity pool, paired with a quote-token deposit in the prevailing reserve ratio.	Own
<i>Withdrawal (USD)</i>	Quote-token amount withdrawn from the AMM liquidity pool (USD) as part of a paired liquidity removal.	Own
<i>Withdrawal (RWA token)</i>	RWA-token amount withdrawn from the AMM liquidity pool as part of a paired liquidity removal.	Own
<i>Swap (USD)</i>	Quote-token amount swapped in the AMM (USD, input amount), conditional on a trade executing against the pool.	Own
<i>Swap (RWA token)</i>	RWA-token amount swapped in the AMM (input amount), conditional on a trade executing against the pool.	Own
<i>Issuer liquidity (USD)</i>	USD value of the AMM pool liquidity attributable to the issuer-designated LP address, identified as the pool-creation (first-seeder) wallet across all pools.	Own
<i>LP trading fee revenue (USD)</i>	Liquidity-provider fee paid on AMM trades (0.25% of trade volume), allocated pro rata to LPs by liquidity share (with an additional 0.05% protocol fee to protocol-token holders).	Own
<i>AMM token exchange rate volatility</i>	Standard deviation of log changes in the AMM-implied exchange rate between the RWA token and the quote token.	Own
<i>AMM slippage (%)</i>	Percentage difference between the pool-implied pre-trade price and the realized execution price on the AMM; computed from on-chain swap amounts and reserve states and therefore reflecting trading fees.	Own
<i>P2P bid offer depth (USD)</i>	Aggregate posted bid depth, converted to USD for a given RWA token at the time of the trade (standing willingness to buy).	Own
<i>P2P ask offer depth (USD)</i>	Aggregate posted ask depth, converted to USD for a given RWA token at the time of the trade (standing willingness to sell).	Own
<i>Total P2P offer depth (USD)</i>	Total posted depth, converted to USD for a given RWA token at the time of the trade, summed across bid and ask sides.	Own
<i>Time on market since offer creation (days)</i>	Elapsed time (days) between offer creation and execution (partial or full fill).	Own
<i>Bid and ask offer overlap (days)</i>	Number of days in which the same wallet simultaneously posts at least one bid and at least one ask offer for the same RWA token.	Own
<i>Spread (simple, %)</i>	Simple percentage spread based on the overlapping ask and bid prices, computed as the percentage difference between the ask price and the bid price. The measure can be negative if the selected bid exceeds the selected ask.	Own
<i>Total offer depth (USD)</i>	Two-sided liquidity proxy for a market-maker–token pair, defined as the smaller of average bid offer depth and average ask offer depth during bid–ask overlap periods.	Own

Quasi-arbitrage opportunities and realized quasi-arbitrage trades

<i>AMM → P2P bid</i>	Quasi-arbitrage opportunity implied by cross-mechanism price signals and defined as an economically feasible ordered pair of buying a RWA token via the AMM and selling into the P2P bid. Importantly, these opportunities are inferred from price signals and need not be feasible at economically meaningful scale or implementable for all investors due to execution risk, quantity caps, and whitelisting.	Own
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<i>AMM → Buyback</i>	Quasi-arbitrage opportunity implied by cross-mechanism price signals and defined as an economically feasible ordered pair of buying a RWA token via the AMM and selling via Buyback. Importantly, these opportunities are inferred from price signals and need not be feasible at economically meaningful scale or implementable for all investors due to execution risk, quantity caps, and whitelisting.	Own
<i>P2P ask → AMM</i>	Quasi-arbitrage opportunity implied by cross-mechanism price signals and defined as an economically feasible ordered pair of buying a RWA token via the P2P ask and selling via the AMM. Importantly, these opportunities are inferred from price signals and need not be feasible at economically meaningful scale or implementable for all investors due to execution risk, quantity caps, and whitelisting.	Own
<i>P2P ask → P2P bid</i>	Quasi-arbitrage opportunity implied by within-mechanism P2P offer-book price signals (ask vs bid) and defined as an economically feasible ordered pair of buying a RWA token via the P2P ask and selling into the P2P bid. Importantly, these opportunities are inferred from price signals and need not be feasible at economically meaningful scale or implementable for all investors due to execution risk, quantity caps, and whitelisting.	Own
<i>P2P ask → Buyback</i>	Quasi-arbitrage opportunity implied by cross-mechanism price signals and defined as an economically feasible ordered pair of buying a RWA token via the P2P ask and selling via Buyback. Importantly, these opportunities are inferred from price signals and need not be feasible at economically meaningful scale or implementable for all investors due to execution risk, quantity caps, and whitelisting.	Own
<i>P2P ask ⇒ AMM</i>	Realized round-trip quasi-arbitrage sequence defined as a buy via the P2P ask followed by a sell via the AMM in the same RWA token within a fixed window, with strictly positive net profit after fees.	Own
<i>P2P ask ⇒ P2P bid</i>	Realized round-trip quasi-arbitrage sequence defined as a buy via the P2P ask followed by a sell into the P2P bid in the same RWA token within a fixed window, with strictly positive net profit after fees.	Own
<i>P2P ask ⇒ Buyback</i>	Realized round-trip quasi-arbitrage sequence defined as a buy via the P2P ask followed by a sell via Buyback in the same RWA token within a fixed window, with strictly positive net profit after fees.	Own
Dependent variables (intraday perspective)		
<i>AMM trade</i>	Indicator equal to one if the trade executes via AMM.	Own
<i>P2P trade</i>	Indicator equal to one if the trade executes via P2P.	Own
<i>Buyback trade</i>	Indicator equal to one if the trade executes via Buyback.	Own
Dependent variables (daily perspective)		
<i>External liquidity (USD)</i>	USD value of the AMM pool liquidity not attributable to the issuer-designated LP address (i.e., provided by non-issuer LPs).	Own
Stress event variables-intraday perspective (±100-days around the shock)		
<i>1-token price impact (bps)</i>	Implied execution cost, in basis points, of a hypothetical 1-token trade (\$50). For AMMs, computed from contemporaneous pool states under constant-product pricing as $10,000 \times \left(1 - \frac{X}{X+1}\right)$, where $X = L^{\text{pool}} / P^{\text{spot}}$ denotes the implied base-token reserve inferred from total pool value and the AMM spot price. For P2P trades, defined as $10,000 \times (P^{\text{exec}} - P^{\text{ask}}) / P^{\text{ask}}$, using the contemporaneous liquidity-weighted ask quote.	Own
<i>P2P bid–ask spread (bps)</i>	Implied cost of immediacy in peer-to-peer trading, expressed in basis points. Computed from contemporaneous bilateral quotes as $10,000 \times (P^{\text{ask}} - P^{\text{bid}}) / P^{\text{mid}}$, where $P^{\text{mid}} = \frac{1}{2}(P^{\text{ask}} + P^{\text{bid}})$ and P^{ask} and P^{bid} denote the liquidity-weighted ask and bid quotes prevailing at the time of observation. The measure captures trading costs conditional on two-sided P2P liquidity.	Own
<i>Asset size (USD)</i>	Concurrent USD appraisal value of the underlying real estate investment at the time of trade, measured by the total investment value associated with the traded token. The variable reflects the contemporaneous fair value used in issuer Buyback and serves as a proxy for underlying asset size. We use the token-level median of this measure to partition the sample into small and large assets in cross-sectional analyses.	Own
<i>TRO affected</i>	Indicator variable equal to one if a trade involves a RWA token directly affected by the Temporary Restraining Order (TRO), and zero otherwise. On July 22, 2025, a Detroit judge granted the City of Detroit a TRO requiring rental collections on affected properties to be halted until certificates of compliance are obtained. As a result, rental income distributions to investors were halted and redirected to escrow accounts, and the Buyback was suspended by the issuer for all affected properties.	Own
<i>Total fees (USD, wavg)</i>	Token–day average of <i>Total fees (USD)</i> across all trades in that token on that day, weighted by <i>Trade volume (USD)</i> .	Own
Market control variables (daily perspective)		
<i>RWA token price volatility (7-day)</i>	7-day rolling standard deviation of daily volume-weighted average prices for a given RWA token.	Own
<i>Market-to-appraisal ratio</i>	Ratio of total RWA market capitalization based on 7-day rolling volume-weighted average prices to total RWA market capitalization based on concurrent Buyback prices.	Own

<i>Cumulative return ETH (one week before)</i>	Cumulative return of Ether over the prior week.	Coinbase
<i>Average ETH tx fee (USD)</i>	Average Ethereum transaction fee over the observation period, expressed in USD.	Coinbase
<i>1m Treasury yield Δ</i>	Daily log-change of the market yield on US Treasury securities at 1-month constant maturity (investment basis).	FRED
<i>10y Treasury yield Δ</i>	Daily log-change of the market yield on US Treasury securities at 10-year constant maturity (investment basis).	FRED
<i>ADS Index Δ</i>	Daily log-change of the Aruoba–Diebold–Scotti (ADS) Business Condition Index (Aruoba et al., 2009).	Fed Philadelphia
<i>S&P Case-Shiller Index Δ</i>	Monthly log-change of the US S&P Case-Shiller National Home Price Index, lagged by one month.	FRED
Trader heterogeneity		
<i>All trader wallets</i>	Seller wallets with at least one observed secondary-market trade in the sample.	Own

Note: This table lists all variables, their definitions, and data sources. *Own* denotes authors’ own calculations based on on-chain transaction records.

A.4. Data in detail

This appendix summarizes the construction and processing of our on-chain datasets for the RealT’s tokenized RWA market. All steps are implemented in Python and R. The full codebase, APIs, and intermediate data outputs are included in the replication package.

A.4.1. Data description

Our primary dataset is a complete transaction-level panel of all secondary-market trades in RealT-issued RWA tokens from Nov. 4, 2019 to Oct. 16, 2025, covering AMM, P2P, and Buyback activity. It contains 777,038 trades (\$41.2 million) across 9,202 unique wallets.² Summary statistics are reported in Tables 1 and 2. We merge trades with token- and property-level fundamentals for 752 tokenized RWAs: 728 are equity claims on SPVs holding one or multiple residential rental properties in the US and Latin America, and the remainder are property-loan tokens, factoring tokens, and tokenized company shares. Overall, the sample covers 923 properties and 1,794 residential units.

A.4.2. Data sources and extraction

Our dataset integrates on-chain event data stored on Ethereum mainnet and Gnosis Chain, whose sources are explained in this subsection.

Identification of relevant smart-contract addresses. We begin by identifying the smart-contract addresses that mediate trading on each marketplace. Starting from the RWA token contract addresses shown on RealT’s primary-market listing pages (RealT Website), we retrace transactions and counterparties using the respective blockchain explorers, Etherscan for Ethereum mainnet and Gnosisscan for Gnosis Chain. To further validate contract identification, one author executed small test trades (below \$100) on each venue and retraced the resulting on-chain interactions. These steps yield the recurring *interaction-contract addresses*, i.e.,

²Multiple wallets may belong to the same individual, implying potential wash trading (Cong et al., 2023).

the smart-contract addresses that traders’ transactions call when executing trades on a given venue, listed below:

- **P2P.**

- Swapcat: 0xB18713Ac02Fc2090c0447e539524a5c76f327a3b (Ethereum and Gnosis)
- YAM: 0xc759aa7f9dd9720a1502c104dae4f9852bb17c14 (Ethereum and Gnosis)

- **AMM.**

- Levinswap: 0xb18d4f69627F8320619A696202Ad2C430CeF7C53 (Gnosis only)
- Uniswap pools do not have a single protocol-level interaction-contract address; events are emitted by the pool (pair) contracts.

- **Buyback.** Buybacks execute directly against the issuer-controlled, token-specific vault (property vault), not a centralized protocol contract. We identify each vault by retracing the initial issuance flow: the earliest transfers originate from the zero address (0x0...0), and the first recipient that subsequently distributes tokens in systematic primary-market sales is the vault. We then classify transfers by direction relative to the vault: OUT (vault→investors) as primary-market sales and IN (investors→vault) as issuer repurchases (Buybacks). We validate this mapping by reconciling cumulative OUT flows with issued supply, accounting for RealT’s standard issuer retention (5%) and excluding rare cases where the issuer supplies inventory directly to secondary venues (STO-like sales). As a further consistency check, we use RealT’s main operating wallet (0x5Fc96c182Bb7E0413c08e8e03e9d7EFc6cf0B099) to confirm that the identified vault engages in the expected issuance and cash-management interactions.

These central addresses serve as fixed reference points throughout our data extraction pipeline and are used for manual cross-checks at each step (contract identification, event retrieval, and test reconstruction outputs using alternative methods involving Etherscan and Gnosisscan APIs).

Indexed on-chain data. To retrieve transaction- and state-level data at scale, we query official RealT and community-maintained subgraphs on The Graph, which index raw blockchain logs into structured entities. The subgraphs used in this paper are maintained under a common creator address 0x1bcfe8666cbb7edd3eda5d343d5f2d4ce853b034. We access each subgraph listed below via [The Graph](https://gateway.thegraph.com/api/[api-key]/subgraphs/id/[subgraph-id]) using the endpoint `https://gateway.thegraph.com/api/[api-key]/subgraphs/id/[subgraph-id]`. An API key can be created free of charge by connecting a wallet to The Graph, and the free-tier rate limits are sufficient to reproduce our full extraction.

Marketplace and subgraph IDs (by blockchain).

- **P2P**

- Swapcat: 4zYp9FYpdepqxqgMCwDTawDvfFbS9VsXmocyiSFLwEvi7 (Ethereum)
- Swapcat: BC5uHuHf4YybNfQECjvHJyZmEfLrPSUu8fLNMZsQ38jx (Gnosis)
- YAM: gMUyhime7v5vkmCv8uJV3L8L3MnLQx2BjTjtIbH6zhk (Ethereum)
- YAM: 2RHalwFYozfHN2bbwUmAoQeJuGn7FX5UREGckZFwfiZi (Gnosis)

- **AMM**

- Levinswap: 6fus6wPYtbNAXr5ZwTrP7anuPypFkEw4snyWFXWbSbAj (Gnosis)
- Uniswap V2: 2zkYvVc8xsmytcEjE9pr523qFcbvqRGxhuGGtraozs66 (Ethereum)
- Uniswap V1: ESnjgAG9NjfmHypk4Huu4PVvz55fUwpyrRqHF21thoLJ (Ethereum, not RealT-curated)

Data validation. We validate indexed records in multiple ways. First, we manually cross-check selected indexed transactions against multiple blockchain explorer outputs, including Etherscan, Gnosisscan, and Blockscout (as an alternative for both) and compare and recompute key decoding logic (e.g., offer enumerations for P2P offer-books, AMM pool-reserve transitions, and implied pool prices). Second, we cross-check subgraph indexer consensus using The Graph’s [Query Cross Checking Tool](#) and rerun identical queries to verify stability across indexers and across repeated calls. Third, at the time of collection, the relevant subgraphs were served by multiple indexers, mitigating single-indexer failures. Finally, for AMMs, we independently recompute pool reserve transitions from swap events and retrace selected liquidity-provider balances and fee accrual logic for spot checks. We further cross-check our results in aggregate with alternative analytical and smart-contract interface applications for RealT’s RWA tokens ([RealT Lab](#)) and RealT as a platform itself (e.g. [DefiLlama](#)).

Further data extraction methods. For the Buyback transactions we use the Etherscan and Gnosisscan APIs for Gnosis and Ethereum and fetch all associated ERC-20 transfers with the corresponding property vault addresses.

A.4.3. Event-level construction and enrichment

Event retrieval and schema. Using the subgraphs listed above, we query all market-place events at the block level. For each event, indexed records include the event type (e.g., mint, swap, burn for AMMs; created, updated, purchase, removed for P2P), transaction hash, log index, block number, timestamp, sender and receiver addresses, token symbols and addresses, and traded quantities. Mechanism-specific fields include pool identifiers, LP fee parameters, LP token holder identifiers and positions (AMMs), and offer identifiers and offer-state variables (P2P).

State reconstruction. For P2P, the event history allows reconstruction of the full offer-book dynamics by offer ID, including updates, partial fills, and removals. For AMMs, we reconstruct liquidity-pool state trajectories, including reserve transitions and implied prices,

as well as liquidity provision and withdrawal activity and the corresponding LP identities at the pool level.

AMM liquidity provision and external participation. We enrich our AMM event data with on-chain liquidity information from three The Graph subgraphs mentioned above: Uniswap V1 (Ethereum), Uniswap V2 (Ethereum), and LevinSwap (Gnosis Chain). Liquidity providers (LPs) receive pool-specific LP tokens when adding liquidity and surrender them when withdrawing liquidity. These LP tokens represent pro-rata claims on pool reserves and trading fees. We separate issuer-provided from externally provided liquidity by defining issuer-related addresses:

- **Issuer-related LP addresses.** We define issuer-related addresses based on (i) repeated pool seeding (earliest `mint` recipient per pool), (ii) involvement in protocol/administrative and buyback-related flows, and (iii) persistent linkages in LP-token transfers around the Ethereum→Gnosis migration 2021. The issuer-related set mainly includes:
 - RealT main pool seeder (> 95% of RWA pools):
 - * 0x5fc96c182bb7e0413c08e8e03e9d7efc6cf0b099
 - Exclusive Uniswap V1 seeder: 0x387ca78a5edf943d267b9248cce1b0fdb67d0029
 - * Ethereum-Gnosis-migration-linked interim addresses (LP-token transfer linkages):
 - * 0xcba15d7eb8fe219a70df21d3259c3cbbf7c6e525,
 - * 0xf22c2e3475e4a066f4e9f44567c950dd36112d05
 - Protocol operational contract (*LevinMaker*):
 - * 0x6d81dda24b7ff5b4a65039ff15d06a076e018e49

Issuer vs. external liquidity provision. For each pool, we retrieve total LP token supply and LP token balances held by issuer-related addresses. We query these at the pool’s first observed block and at all `mint`/burn blocks, then forward-fill within pool between queried blocks. Issuer liquidity is measured by issuer-held LP tokens; external liquidity is the remainder (total supply minus issuer-held LP tokens). We measure *net liquidity provision* as signed changes in LP token balances at `mint`/burn blocks (positive deposits, negative withdrawals), aggregated by pool-day and split by issuer-related vs. external LPs. Using daily end-of-day snapshots of LP token holders (positive balances) excluding issuer-related addresses, we further report the number of active external LPs and the cumulative number of unique external LPs over time.

Transaction fees and USD conversions. Using transaction hashes, we augment on-chain events with realized network fees from the Etherscan and Gnosisscan APIs. USD conversions use the [Coinbase API](#): ETH-denominated values are mapped to USD using hourly ETH/USD rates. For LEVIN, we infer LEVIN/USD from contemporaneous WXDAI/LEVIN pool reserves via the implied spot price (WXDAI being USD-pegged on Gnosis).

Property-level and token fundamentals We collect token-, asset-, and platform-level fundamentals from the RealT website, the community-maintained [RealT API](#), and RealT’s social media communications. We use token metadata, STO terms, rental-income histories, and appraisals to enrich the on-chain panels with asset characteristics and platform events. Time-invariant attributes include issuance volume, token supply, issuance date, location, and property characteristics. Time-varying attributes include rental income, rent-to-value yields, rental status, building age, and appraisal values.

A.4.4. Sample filters

Our sample spans Nov. 4, 2019, 16:07:31 (Ethereum block 8872285) to Oct. 16, 2025, 08:59:35 (Gnosis block 42651526). Block timestamps are in UTC. To ensure mechanical precision and remove economically implausible records, we further apply the following mechanism-specific rule-based filters.

AMM. We exclude four AMM swaps with implausible implied prices above \$1,000,000 and one swap below \$0.10. These bounds are intentionally conservative to avoid over-filtering in thin pools where deterministic AMM pricing can produce large dispersion. We do not winsorize or otherwise smooth AMM price signals, since extreme quotes are informative about low depth and are reported as such (see Table 2). From AMM subgraphs, we reconstruct liquidity-pool states (reserves, implied prices, execution prices, slippage, and liquidity-provider positions), yielding a pool-level panel.

P2P. We restrict attention to offers and purchases that involve at least one RWA token and are quoted against established stablecoins (USDC, USDT, WXDAI, DAI, REUSD, MAI). We drop all other pairs, including RWA token–RWA token transfers, which removes 636 offers and 349 purchases. Because wallets can post offers exceeding their on-chain balances, we discard offers whose advertised quantity exceeds 30% of total token supply. This threshold is above the 15% primary-market allocation limit and remains permissive of secondary-market accumulation. This removes 10,757 offers and 1,108 purchases. We further exclude offers hard-deleted by RealT administrators during three documented purge blocks on YAM (619 offers and 1,191 purchases) and observations with missing or implausible USD prices outside the \$1–\$1,000 range (1,377 offers and 307 purchases). Together, these filters remove approximately 6.9% of offer events and 1.4% of purchase events, leaving 179,788 offers and 207,885 purchases. Purchases exceed offers because a single offer can be partially filled multiple times, generating multiple purchase records.

Buyback. One Buyback transaction is identified as an internal transfer and excluded; all remaining observations are retained. Buyback transactions are identified by tracing token flows between issuer and external wallets and netting these flows against STO issuance volumes.

A.4.5. Reproducibility

For reproducibility, our replication package includes scripts to rerun the full extraction and reconstruction pipeline. Because 99% of secondary-market trading volume in our sample

settles on Gnosis Chain, we run the Gnosis portion against a local archive node to ensure access to historical state at arbitrary block heights and to avoid reliance on third-party indexer availability. The archive node is queried via a [Chainstack](#) Remote Procedure Call (RPC) endpoint, and all intermediate outputs are written to disk to permit exact reruns and step-wise verification.

A.5. Quasi-arbitrage analysis

Quasi-arbitrage. To reflect market frictions, including execution uncertainty, heterogeneous mechanisms, and appraisal-anchored pricing, we term the exploitation of cross-mechanism price dispersion *quasi-arbitrage* rather than risk-free arbitrage. Panel A of Table A.6 shows that median prices are tightly aligned across mechanisms, while the mean AMM quote and its dispersion are substantially larger, reflecting a small number of very thin pools. Panel B translates these price signal differentials into token-day quasi-arbitrage opportunities for each economically feasible ordered pair of buy and sell mechanisms, denoted as $m^{\text{buy}} \rightarrow m^{\text{sell}}$. Importantly, these opportunities are inferred from contemporaneous price signals rather than executed trades and therefore need not be feasible at economically meaningful scale, nor implementable for all investors given quantity caps, execution risk, and whitelist-based access restrictions. Panel C reports realized round-trip quasi-arbitrage buy-sell sequences, defined as a buy followed by a sell in the same token within a fixed window (1 minute or 1 hour) with strictly positive net profit after fees. We denote such sequences by $m^{\text{buy}} \Rightarrow m^{\text{sell}}$. To capture cross-mechanism price pressure even when trades route through intermediaries or smart-contract routers, Panel C imposes no wallet identity constraint, yielding conservative lower-bound estimates of actual trading activity. Across both horizons, realized quasi-arbitrage is concentrated in $P2P \text{ ask} \Rightarrow \text{AMM}$ and $P2P \text{ ask} \Rightarrow P2P \text{ bid}$ sequences.

Table A.6: Price signals and quasi-arbitrage

<i>Panel A: Summary statistics of price signals</i>				
Mechanism	Token-days	Median (USD)	Mean (USD)	St. Dev. (USD)
Price signal AMM (USD)	187,749	52.67	62.44	5,013.91
Price signal P2P bid offer (USD)	79,931	49.64	51.39	13.46
Price signal P2P ask offer (USD)	158,232	55.00	56.66	11.19
Price signal Buyback (USD)	188,198	50.79	51.14	13.99
<i>Panel B: Quasi-arbitrage opportunities from price signals (token-day level)</i>				
Buy→Sell	Token-days with opportunity	Median spread (USD)	Mean spread (bps)	Median spread (bps)
AMM→P2P bid	7,546	1.34	567.60	258.29
AMM→Buyback	48,426	1.13	385.14	217.91
P2P ask→AMM	45,350	2.09	757.18	362.98
P2P ask→P2P bid	1,897	1.22	418.73	233.39
P2P ask→Buyback	8,567	0.50	285.42	95.69
<i>Panel C: Realized quasi-arbitrage trades (transaction-pair level, no wallet constraint)</i>				
Buy⇒Sell	Token-pairs	Token-quantity	Net-profit (USD)	Net-volume (USD)
<i>1-minute window</i>				
P2P ask⇒AMM	10,399	1,302.12	4,305.86	80,743.22
P2P ask⇒P2P bid	1,752	2,792.58	2,591.28	150,016.17
<i>1-hour window</i>				
P2P ask⇒P2P bid	21,573	41,692.72	106,187.95	2,279,314.96
P2P ask⇒AMM	63,590	3,489.56	11,215.12	204,863.33
P2P ask⇒Buyback	42	46.02	312.07	3,088.01

Note: Panel A reports token-day coverage (Token-days) and the median, mean, and standard deviation computed across transaction-level signal observations on those token-days. Panel B reports token-day quasi-arbitrage opportunities implied by these signals and denoted by $m^{\text{buy}} \rightarrow m^{\text{sell}}$ (buy via m^{buy} , sell via m^{sell}). Panel C reports realized quasi-arbitrage trades denoted by $m^{\text{buy}} \Rightarrow m^{\text{sell}}$ and defined as a buy followed by a sell in the same token within the indicated time window, with strictly positive net profit after fees; net volume is the sell-leg notional at net prices. Please note that, especially for P2P, many realized sequential trades occur within a single day, relative to the token-day level of analysis in Panel B. The AMM price signal is identical for bid and ask by construction, so we do not distinguish between them in our notation. The Buyback mechanism provides only a bid quote, as tokens can only be sold into it.